



# Intrinsic Arm Exponents in High-Dimensional Percolation

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# Chapter 1

## Introduction

Percolation describes the process of a liquid, gas, etc. moving randomly through a medium that has very small holes or spaces in it, which the non-solid substance can use to spread. With repeated observations the result should look slightly different every time and we get a random object. There are two ways to model this. We could assign the random mechanism to the fluid, meaning the medium is fixed and the fluid spreads out randomly described by a diffusion process. Another strategy would make the connections inside the medium random, so we would model a stochastic medium. In this work we will examine the latter model: Imagine having a porous stone or a chunk of dirt that we throw into a pool of water. Since the surface of the medium is not waterproof, some portion of the solid material will get wet eventually. But what is the exact fraction that will get wet? One will not get the same result for every porous stone, because some might be more penetrable and others more solid. The mathematical model to analyze this was first introduced by Broadbent and Hammersley [7] in 1957 and over the decades a rich theory was developed. Topics of percolation are now an important research subject of modern probability theory. We will recognize that the behavior of the model is substantially different if we change the connection properties of our medium. This is why percolation reckons to be one of the simplest models displaying critical phenomena and is a paradigm model of statistical physics.

The main question of this study is, how the probability behaves that inside our medium we observe a shortest path of a certain length. If we start at a given position and only use the shortest connection, how far can we travel and in what way does this depend on the desired distance? This is already hidden in the thesis title, which raises some questions to one who is unfamiliar with the used vocabulary: What is an intrinsic arm exponent? What exactly are high dimensions? What precisely is percolation and how do we describe it mathematically? We will try our best to give a satisfying answer to these questions with the following outline:

- In Chapter 2 we give a short introduction to the most basic percolation model and provide an intuition for the theory we will use. Both the definition of the model and the main theorems are given. We claim that in regard to results on percolation that we need, this thesis is self contained. The reader should have profound knowledge in probability theory and elementary mathematics but no prior knowledge of percolation is expected. For readers who are already familiar with percolation, this section can be used as a short revision of the theory.

- After that we turn our head to results that are specific to the high-dimensional case. We will see why we want to study the model in high dimensions and establish the main results in this case. To do so, mean-field behavior is explained and critical exponents are defined.
- With useful theory at hand, Chapter 4 then gives an overview over the intrinsic one arm exponent that we define there. To achieve this we will give proofs for the hypercubic lattice and the proof of the corresponding mean-field value. We then discuss further current research progress and give a short excursion to the related notion of the extrinsic arm exponent.
- To round this up, we then study the occurrence of multiple shortest paths of certain length and introduce the theory regarding magnetization and ghost fields. We define the intrinsic  $l$ -arm exponent, give mean-field bounds on it and state a conjecture. Two promising proof strategies to tackle this problem are also discussed.

# Chapter 2

## Preliminaries

In this chapter we will give a short introduction to bond percolation on  $\mathbb{Z}^d$ . We go over all the theory needed later, but omit the proofs. We will have to skip some topics that would be normally covered in a textbook, therefore this is not a complete introduction to percolation. For that we can recommend the lecture notes of Duminil-Copin [12], wherein one may also find suitable exercises, which is always useful since mathematics is not a spectator sport. If a broader and more detailed coverage is desired we can give the book of Grimmett [18] as a reference. For more general graphs [37] can be consulted.

In percolation theory we interpret  $\mathbb{Z}^d$  as a graph  $(\mathbb{Z}^d, \mathcal{E}(\mathbb{Z}^d))$ , also known as the  $d$ -dimensional hypercubic lattice. Elements of  $\mathbb{Z}^d$  are then called *vertices*, *points* or *sites*. With  $\mathbb{Z}^d$  we will refer to both the graph and the vertex set. The edges are also called *bonds*. For our edge set we want to connect vertices if they are nearest neighbors in the graph, so  $\mathcal{E}(\mathbb{Z}^d) = \{\{x, y\} \subset \mathbb{Z}^d \mid |x - y| = 1\}$ , where  $|\cdot|$  denotes the euclidean norm

$$|x| = \|x\|_2 = \left( \sum_{i=1}^d x_i^2 \right)^{\frac{1}{2}}, \quad x \in \mathbb{Z}^d. \quad (2.1)$$

For  $x, y \in \mathbb{Z}^d$  we write  $x \sim y$ , if  $x$  and  $y$  are neighbors and say that they are *adjacent*. This is then called the *nearest-neighbor model*. We could also put edges between  $x$  and  $y$  if  $|x - y| \leq L$ , where  $L > 0$  is a typically very large number. This is then called the *spread-out model*. However we will focus on the classical nearest-neighbor model.

Studying only a fixed graph may be a bit boring. Hence we now introduce some probability to this graph. With probability  $0 \leq p \leq 1$  we declare each edge independently to be *open/occupied*. If edges are not open, they are called *closed/vacant*. Thus we sample a random subgraph of  $\mathbb{Z}^d$ . As always when studying probabilistic objects, we want to model this with a probability space  $(\Omega, \mathcal{F}, \mathbb{P}_p)$ . A natural good choice for our sample space would be  $\Omega = \{0, 1\}^{\mathcal{E}(\mathbb{Z}^d)}$ . A *configuration*  $\omega \in \Omega$  can then be represented as  $\omega = (\omega(e) : e \in \mathcal{E}(\mathbb{Z}^d))$ , where  $\omega(e) = 1$  stands for an open edge and  $\omega(e) = 0$  for a closed one.

For our  $\sigma$ -Algebra we take the one generated by all cylinder sets, so

$$\mathcal{F} = \sigma(\{C(\omega, K) : \omega \in \Omega, K \subset \mathcal{E}(\mathbb{Z}^d), |K| < \infty\}), \quad (2.2)$$

and for  $\omega \in \Omega$ ,  $K \subset \mathcal{E}(\mathbb{Z}^d)$

$$C(\omega, K) := \{\tilde{\omega} \in \Omega : \forall e \in K \ \omega(e) = \tilde{\omega}(e)\}, \quad (2.3)$$

denotes the  $K$ -cylinder of  $\omega$ . So the  $\sigma$ -Algebra is generated by all events depending on finitely many edges.

Finally let  $\mathbb{P}_p$  be the corresponding product measure on  $\{0, 1\}^{\mathcal{E}(\mathbb{Z}^d)}$ , where each coordinate is a Bernoulli random variable with parameter  $p$ . We label the expectation with respect to  $\mathbb{P}_p$  by  $\mathbb{E}_p$ . It can be shown that the measure  $\mathbb{P}_p$  is translation invariant and ergodic. For this we remember that an event  $A$  is *invariant under translation*, if for all  $x \in \mathbb{Z}^d$  we have  $\tau_x A = A$ , where  $\tau_x$  shifts the lattice by  $x$ . This means for a set we have  $\tau_x A = \{\omega \in \Omega : \tau_x^{-1} \omega \in A\}$ . A measure is now said to be *ergodic* if events that are invariant under translation either have probability 0 or 1 and nothing in between. Additionally a measure  $\mu$  is *invariant under translation*, if  $\mu(\tau_x A) = \mu(A)$  for any event  $A$  and  $x$ .

We are foremost interested in the analysis of connected components. For that we write  $x \longleftrightarrow y$  if there exists a *path* connecting  $x$  and  $y$ . This means there are  $x = v_0, v_1, \dots, v_n = y \in \mathbb{Z}^d$  with  $n \in \mathbb{N}$ , such that  $\{v_{i-1}, v_i\} \in \mathcal{E}(\mathbb{Z}^d)$  and  $\omega(\{v_{i-1}, v_i\}) = 1$  for all  $i = 1, \dots, n$ . We then use the shorter notation  $\{x \longleftrightarrow y\} = \{\omega \in \Omega : x \longleftrightarrow y \text{ on } \omega\}$ .

If we explore all the sites connected to  $x$ , we get the *cluster* of  $x$

$$\mathcal{C}(x) := \{y : x \longleftrightarrow y\}. \quad (2.4)$$

By convention  $x \longleftrightarrow x$  always holds, so  $x \in \mathcal{C}(x)$ . Next we define one of the most important objects in percolation theory. The *percolation function*  $p \mapsto \theta(p)$  is the probability that the origin is inside an infinite cluster, thus

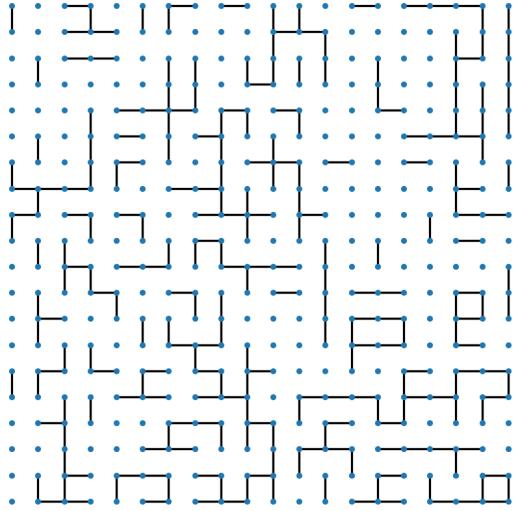
$$\theta(p) := \mathbb{P}_p(|\mathcal{C}(0)| = \infty). \quad (2.5)$$

Note that we slightly abuse the notation here, since  $|\cdot|$  is now obviously the cardinality of the set and not a norm. Picking  $0 \equiv (0, \dots, 0) \in \mathbb{Z}^d$  as the origin and our reference point is arbitrary. Also translation invariance implies that it is irrelevant which specific point we choose, since the probability of being inside an infinite component is the same for all sites. See Figure 2.1 for a visualization of a sampled cluster, the pictures should make the model more lucid.

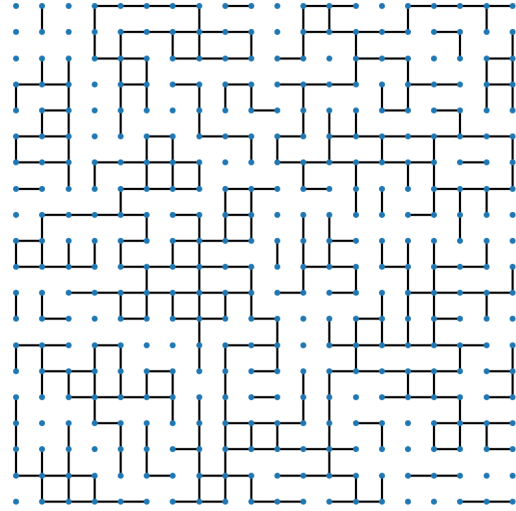
In percolation, classical questions would be when and how many infinite clusters occur. For this we define the *critical value* as a central object which acts as a threshold. This critical value is now

$$p_c = p_c(\mathbb{Z}^d) := \inf\{p : \theta(p) > 0\}. \quad (2.6)$$

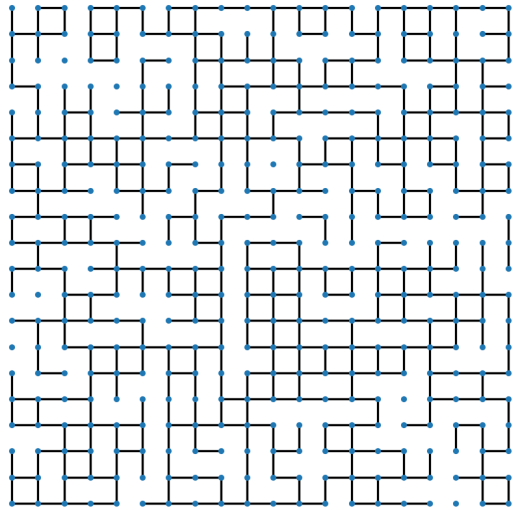
For  $d > 1$  the value is non trivial in the sense that  $0 < p_c < 1$ . Indeed it is straightforward that on a line  $p_c(\mathbb{Z}) = 1$ , since two vacant edges on each side can easily block an infinite structure. This is not the case for higher dimensions, where in general one closed edge cannot fully separate the graph. Therefore we will not pay a lot of attention to the one dimensional case. The value of  $p_c$  is of immense importance, since the behavior of the model drastically changes around  $p_c$ . This is why we also say that percolation has a non trivial *phase transition*. For values  $p_c > p$  we are in the *subcritical* region,  $p_c = p$  is *critical* and for  $p_c < p$  we speak of *supercritical* percolation. For dimension  $d = 2$  the critical value was computed explicitly by Kesten in [30], where  $p_c(\mathbb{Z}^2) = 1/2$  was proven.



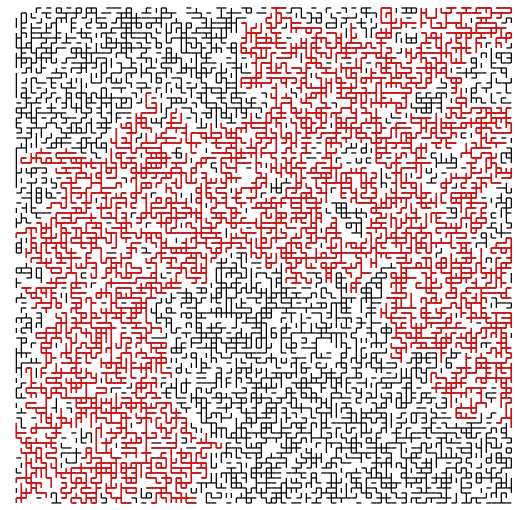
(a) Subcritical:  $p = 1/3$



(b) Critical:  $p = 1/2$



(c) Supercritical:  $p = 2/3$



(d) Critical percolation on a  $100 \times 100$  lattice. The cluster of the origin (for visualization purpose the corresponding edges) is colored red.

Figure 2.1: Some examples of percolation on a  $20 \times 20$  grid for different values of  $p$  and one more zoomed out picture with the cluster of the origin colored in red. Unsurprisingly, we observe more connections and thus larger connected components for higher values of  $p$ . The simulations were done using Python with the NetworkX package.

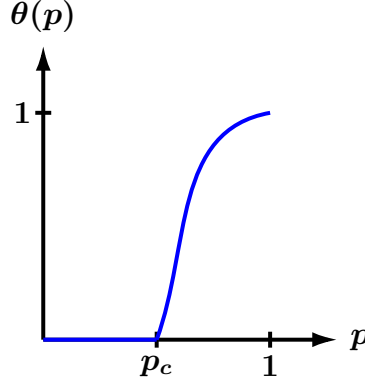


Figure 2.2: Conjectured drawing of the percolation function, so  $\theta(p_c) = 0$ .

A natural thing one might want to study now is the behavior of the percolation function around  $p_c$ . We summarize this in the following lemma. See also Figure 2.2 for a sketch.

**Lemma 2.1** (Behavior of the percolation function [41, 47]). *The following properties hold:*

1.  $\theta(p) = 0$  for  $p < p_c$  and  $\theta(1) = 1$ .
2. For  $p > p_c$  we find that  $\theta(p) > 0$ . Furthermore in the supercritical regime the percolation function is continuous, increasing and infinitely many times differentiable.
3.  $\theta(p)$  is right continuous at  $p_c$ .

It is clear from the definition that  $\theta(p) = 0$  for  $p < p_c$  and  $\theta(1) = 1$ . Since in the former case  $\mathbb{P}_p(|\mathcal{C}(x)| = \infty) = 0$  for all sites  $x$ , we have no infinite cluster almost surely.

It would be advantageous if the percolation function was also left continuous at  $p_c$ . Is this the case? It is conjectured that for all  $d \geq 2$  we have  $\theta(p_c) = 0$ . For  $d = 1$  the claim would be wrong, since there  $\theta(p_c) = 1$ . Note that  $\theta(p_c) = 0$  also implies that the percolation function is continuous on the whole domain. For dimension  $d = 2$  this conjecture was resolved by Kesten [30]. The claim was also shown for  $d > 10$  and we will later elaborate on this in Chapter 3. For dimensions in between the conjecture is completely open. This can be seen as the most famous problem in percolation theory and it seems unreachable (in full generality) with today's methods. If we use a search engine to look for big open problems in modern probability theory, this conjecture will show up quickly. We will see that it is typical for percolation on the hypercubic lattice, that we have rich results for dimension  $d = 2$  and in high dimensions, yet lack rigorous results in between.

Now we have a rough idea how the probability that a site is inside an infinite cluster behaves. We could also ask how many infinite clusters are present in the supercritical region, if we look at a fixed configuration. For  $\theta(p) > 0$  we can use Kolmogorov's zero-one law (clearly the existence of an infinite component is a tail event) and follow that we have at least one infinite cluster almost surely. An alternative proof would use the ergodicity of  $\mathbb{P}_p$  and that the existence of an infinite component is translation invariant. It is also easy to see that the only two possibilities are one unique cluster or infinitely many clusters. It was first shown in [2] that the former is true, so in the supercritical region we have one unique infinite component almost surely. We could further be interested in the expected size of the cluster of the origin. For this define the *susceptibility*  $\chi(p)$  as

$$\chi(p) := \mathbb{E}_p[|\mathcal{C}(0)|]. \quad (2.7)$$

It is clear that in the supercritical region  $\chi(p) = \infty$ , since the event  $\{|\mathcal{C}(0)| = \infty\}$  has a positive weight assigned to it. As we saw, we have a phase transition around  $p_c$ , but we could define another threshold where the susceptibility diverges. The following shows that these values coincide.

**Theorem 2.2** (Unique phase transition [1, 14, 38]). *If  $p < p_c$ , the susceptibility is finite. At criticality  $\chi(p_c) = \infty$  holds.*

Now we will introduce the arguably most important tool in percolation theory: the *BK inequality*. The abbreviation was chosen in honor of van den Berg and Kesten [48], who first proved it in 1985.

Note that we can define a partial order  $\leq$  on  $\Omega$ , if we compare all edges pointwise. So write  $\omega \leq \tilde{\omega}$  if  $\omega(e) \leq \tilde{\omega}(e)$  for all  $e \in \mathcal{E}(\mathbb{Z}^d)$ . A function  $f : \Omega \rightarrow \mathbb{R}$  is then called *increasing* if  $\omega \leq \tilde{\omega}$  implies that  $f(\omega) \leq f(\tilde{\omega})$ . Likewise an event  $A \in \mathcal{F}$  is increasing if the function  $\omega \mapsto \mathbb{1}_A(\omega)$  is increasing and similarly decreasing if  $A^c$  is increasing. A bit informally one might think of events where adding edges is “good” for the event. An example for an increasing event would be  $\{x \longleftrightarrow y\}$  or  $\{|\mathcal{C}(0)| = \infty\}$ . Surely adding edges in these cases is advantageous. On the other side the existence of a unique path would not be increasing. We next define a new operator that combines two events. For a configuration  $\omega$  inside an event  $A$  we say that  $I \subset \mathcal{E}(\mathbb{Z}^d)$  is a *witness of  $A$  for  $\omega$*  if any other configuration  $\tilde{\omega}$  coinciding with  $\omega$  on  $I$  is also in  $A$ . For events  $A, B$  we write  $A \circ B$  for the set of all configurations  $\omega$ , where there exist two disjoint witnesses  $I = I(\omega)$  and  $J = J(\omega)$  of  $A$  and  $B$  for  $\omega$ . We call  $A \circ B$  the *disjoint occurrence of  $A$  and  $B$* . Intuitively, this means that both events hold ( $A \circ B \subseteq A \cap B$  is always true), but they use different edges. It is crucial, that witness sets can be different for distinct configurations.

The observation that the probability of  $A \circ B$  given  $B$  is less than the probability of  $A$  is reasonable, since the event  $B$  already uses some edge that  $A$  might need. The next theorem makes this formal.

**Theorem 2.3** (BK inequality [48]). *If  $A$  and  $B$  are increasing events, then*

$$\mathbb{P}_p[A \circ B] \leq \mathbb{P}_p[A]\mathbb{P}_p[B]. \quad (2.8)$$

In the original work, van den Berg and Kesten conjectured that the above statement can also be formulated more generally for sets that are not necessarily increasing. This was shown by Reimer [40].

**Theorem 2.4** (BK-Reimer inequality [40]). *If  $A$  and  $B$  are events depending on finitely many edges, then*

$$\mathbb{P}_p[A \circ B] \leq \mathbb{P}_p[A]\mathbb{P}_p[B]. \quad (2.9)$$

It would make a lot of sense if for two increasing events  $A$  and  $B$  we have

$$\mathbb{P}_p(A \mid B) \geq \mathbb{P}_p(A), \quad (2.10)$$

so  $A$  and  $B$  are positively correlated. This should hold, because if  $B$  occurs we hope that  $A$  can use the edges which are open for  $B$  to hold. This claim follows from the FKG inequality, sometimes named after Harris, who first proved it in 1960. Later on it was verified in greater generality by Fortuin, Kasteleyn and Ginibre.

**Theorem 2.5** (FKG inequality [16, 22]). *For two increasing functions  $f$  and  $g$  in  $L^2(\Omega)$*

$$\mathbb{E}_p[fg] \geq \mathbb{E}_p[f]\mathbb{E}_p[g]. \quad (2.11)$$

Hence for two increasing events  $A$  and  $B$

$$\mathbb{P}_p(A \cap B) \geq \mathbb{P}_p(A)\mathbb{P}_p(B). \quad (2.12)$$

Another result we want to use later on will be Russo's formula. The classical version of this normally deals with increasing events, but we will need a more general version.

**Theorem 2.6** (Russo's formula [42]). *Let  $X$  be a random variable which depends on finitely many edges. Then  $\mathbb{E}_p X$  is a polynomial in  $p$  with derivative*

$$\frac{d}{dp} \mathbb{E}_p X = \sum_{e \in E} \mathbb{E}_p [X(\omega^e) - X(\omega_e)] = \frac{1}{p} \sum_{e \in E} \mathbb{E}_p [\mathbb{1}(\omega(e) = 1)(X(\omega) - X(\omega_e))], \quad (2.13)$$

where  $\omega^e$  is the configuration  $\omega$  with the edge  $e$  being occupied and  $\omega_e$  is the configuration  $\omega$  with the edge  $e$  being vacant.

With the FKG inequality, BK inequality and Russo's formula we now have the three most important tools at hand for percolation theory.

More generally, we could also perform percolation on any other graph  $G = (V, E)$ . For later results, we say that a graph is (vertex) *transitive* if its automorphism group acts transitively on its vertices. This is just a complicated way of saying that for any  $v_1, v_2 \in V$  there exists a function  $f : V \rightarrow V$  such that  $f(v_1) = v_2$  and  $\{f(x), f(y)\} \in E$  if and only if  $\{x, y\} \in E$  for all  $x, y \in V$ . Rather informally, this means that vertices look alike. Likewise, a graph is *quasi-transitive* if its vertex set can be partitioned into finitely many sets and in the above construction  $v_1$  and  $v_2$  are then chosen only from the same subset. Here, the intuition is that we have a finite number of different vertex types.

Since  $\mathbb{Z}^d$  has a nice euclidean embedding, it is the standard graph for percolation and we will therefore focus on this case. If we present results on more general graphs it will be explicitly stated. Otherwise, we always refer to  $\mathbb{Z}^d$ .

We could also do the same probabilistic procedure for the sites, i.e. we could color sites with probability  $p$ . This would then be called *site percolation*. It can be shown that every bond percolation model can be translated to site percolation (on a potential different graph). If we have a bond percolation model on  $G$ , we can define an equivalent site percolation model on  $G_s = (V_s, E_s)$  as follows. Take  $V_s$  to be the edges of  $G$  and add  $\{a, b\} \in E_s$  if  $a$  and  $b$  share a common endpoint in  $G$  (be careful:  $a$  and  $b$  are sites in  $G_s$ , but edges in  $G$ ). With this construction we have transformed bond percolation to site percolation, and the cluster of  $v \in V$  is infinite if there exists  $e \in E$  such that the corresponding component of  $e \in V_s$  is infinite. The opposite statement is false: Not every site percolation model can be seen from a bond percolation model perspective. Even though site percolation seems to be more general, in this thesis we focus on bond percolation, since the majority of the literature we rely on does the same.

# Chapter 3

## High-Dimensional Percolation

We saw that percolation on  $\mathbb{Z}^d$  has two defining parameters: the dimension  $d$  of our lattice and the probability  $p$ , which is tuning the occurrence of open edges. The title of this thesis implies that we are in particular interested in high dimensions. This might already tell us that percolation in this case is somewhat special or different. We should always keep in mind that  $p_c = p_c(\mathbb{Z}^d)$  is dependent on the graph, or in our case on the dimension  $d$ . We saw that  $p_c(\mathbb{Z}^2) = 1/2$ , but how is  $p_c$  behaving if we increase the dimension? Intuition tells us that the critical probability should be decreasing in dimension. This is because adding more edges makes big clusters more plausible, if we keep the probability of them being open the same. Thus the infimum in the definition of  $p_c$  will be lower. The next lemma makes this more quantitative.

**Lemma 3.1** (Bounds on  $p_c$  [21]). *For percolation there exist  $d_0 > 6$  and  $C_{p_c} > 0$  such that for  $d \geq d_0 > 6$*

$$1 \leq 2dp_c \leq 1 + \frac{C_{p_c}}{d}. \quad (3.1)$$

The proof of the lower bound uses a path counting argument and is true for every dimension  $d$ . Certainly, if the origin is inside an infinite cluster, we find self avoiding walks with arbitrary length starting in 0. For that, let  $S_n$  be the set of all self-avoiding paths starting in 0. For each  $n \geq 1$  we can use subadditivity and write

$$\begin{aligned} \theta(p) &\leq \mathbb{P}_p(\text{there exists a path } \gamma \in S_n \text{ s.t. } w(e) = 1 \text{ for all } e \in \gamma) \\ &\leq \sum_{\gamma \in S_n} \mathbb{P}_p(w(e) = 1 \text{ for all } e \in \gamma) = |S_n| p^n. \end{aligned} \quad (3.2)$$

We can estimate the number of self avoiding walks of length  $n$  by the number of arbitrary walks of length  $n$ , so  $(2d)^n$ . Therefore for all  $n$

$$\theta(p) \leq (2dp)^n \quad (3.3)$$

and the upper bound converges to 0 for  $n \rightarrow \infty$  if and only if  $p < 1/2d$ . We follow that  $p_c \geq 1/2d$ . The bound can even be improved to  $p_c \geq 1/(2d - 1)$ , if we are more careful when estimating  $|S_n|$  and can be generalized to all transitive graphs. From the upper bound of (3.1) we can follow that  $\lim_{d \rightarrow \infty} p_c = 0$ , as expected. The right hand side of (3.1) also gives us control over the expected number of neighboring vertices connected by an open bond. The expectation is obviously  $2dp_c$ , since we have  $2d$  possible connections

and every one is present independently with probability  $p_c$ . Thus the number of expected adjacent vertices is decreasing in dimension to 1, even though with higher dimension we have more possible neighbors. This might be a first hint that we do not expect many circles or loops in high dimensions.

Sometimes in mathematics things can get more complicated when we increase the dimension. Maybe we have more parameters, a lot of indices that might confuse us, or visualization is harder. Yet this does not necessarily imply that we have a worse understanding of the model. We will see in the following that for high dimensions we have very rich results and to some extent a superior perception of the model compared to the lower dimensional case. We already briefly foreshadowed this in the case of the percolation function in Chapter 2. Our main reference for the upcoming results will be [25].

### 3.1 Critical Exponents

We saw that the behavior of the model changes drastically around  $p_c$ . We are therefore predominantly interested in results at  $p_c$  or close to criticality. To tackle this we want to have *power law* expressions for numerous objects. The power is then called a *critical exponent*. Notions like these are often best explained with examples. For instance we predict that there is some dimension dependent  $\beta > 0$ , such that

$$\theta(p) \asymp (p - p_c)^\beta, \quad \text{as } p \searrow p_c. \quad (3.4)$$

Here, the sign  $\asymp$  needs some explanation. There are many different meanings of this in the literature and we are lucky that we only need one case. We write  $\asymp$  if there exist constants  $0 < c_\beta < C_\beta < \infty$ , such that for every  $p \in (p_c, 1]$

$$c_\beta(p - p_c)^\beta \leq \theta(p) \leq C_\beta(p - p_c)^\beta \quad (3.5)$$

and say that  $\beta$  exists in the *bounded-ratio sense*. We should always keep in mind that the constants  $c_\beta, C_\beta$  can (and most of the time will) be dependent on the dimension. Sometimes we will omit the limit statement  $p \searrow p_c$  and write  $\asymp$  for the existence of lower and upper bounds like in (3.5). We stress the point that the existence of a critical exponent is nowhere near obvious and needs rigorous proof. A lot of these results were first expected to hold by physicists and then later confirmed by rigorous arguments. The existence of  $\beta$  is a very powerful result since it directly implies  $\theta(p_c) = 0$  and therefore the continuity of the percolation function.

Another important critical exponent can be defined for the *susceptibility*. The critical exponent  $\gamma > 0$  describes the expected cluster size and is defined by

$$\chi(p) \asymp (p_c - p)^{-\gamma}, \quad \text{as } p \nearrow p_c. \quad (3.6)$$

Arguably the most useful exponent is  $\eta \geq 0$ , which describes the decay rate of the *two point function*. For  $1 \leq p \leq 1$  define the two-point function  $\tau_p(x, y)$  by the probability of a connection between  $x$  and  $y$ :

$$\tau_p(x, y) := \mathbb{P}_p(x \longleftrightarrow y) = \mathbb{P}_p(0 \longleftrightarrow (x - y)). \quad (3.7)$$

We expect the two point function to be decreasing for greater distances. Indeed, long range connections should be less likely than connecting very close sites. This thought is described by

$$\tau_{p_c}(0, x) = \tau_{p_c}(x) \asymp |x|^{-(d-2+\eta)}, \quad \text{as } |x| \rightarrow \infty. \quad (3.8)$$

The last exponent we will need expresses the tail decay of the critical cluster:

$$\mathbb{P}_{p_c}(|\mathcal{C}(0)| \geq n) \asymp n^{-1/\delta}, \quad \text{as } n \rightarrow \infty. \quad (3.9)$$

We can easily give a bound on plausible values, if we say that  $\delta \geq 1$  should hold, since we already know that  $\chi(p_c) = \infty$ . This can be seen, if we remember that for a positive integer valued random variable  $X$

$$\mathbb{E}X = \sum_{n \geq 1} \mathbb{P}(X \geq n). \quad (3.10)$$

With this neat formula we can then write

$$\chi(p_c) = \sum_{n \geq 1} \mathbb{P}_{p_c}(|\mathcal{C}(0)| \geq n) \leq C_\delta \sum_{n \geq 1} n^{-1/\delta}, \quad (3.11)$$

and the right hand side is finite if and only if  $\delta < 1$ . We will see that these critical exponents are one of the main reasons why we have a rich theory of percolation in high dimensions. They will carry the difficult parts of our proofs later when we talk about another critical exponent which is indicated by the title of the thesis. We will not define it here but in Chapter 4, since we will discuss it in much greater detail.

These stated critical exponents are also of same importance when we talk about percolation on more general graphs and not  $\mathbb{Z}^d$ , but we will not need the particular results here. It is believed however that these exponents only depend on the dimension of the graph and not on the individual structure of the lattice. The intuition behind this is that critical phenomena only show themselves over very long scales and the local structure is not that relevant in the macroscopic picture. While this universally is definitely extremely interesting and useful because we can restrict ourselves to “easier” models, rigorous proofs for this exist only rarely. For example Grimmett and Manolescu showed that in two dimensions the critical exponents in bond percolation (assuming they exist) for the square, triangular and hexagonal lattice are the same, so these models would belong to the same universality class [19].

## 3.2 Mean-Field Behavior

Readers familiar with self-avoiding random walks already know that in high dimension they behave similar to simple random walks. But even with no previous knowledge, one can make sense of this. When we have a very vast space and a lot of directions our random walk can move, forming a loop or a cycle becomes improbable. The dimension after the behavior of the model changes is often called *upper critical dimension* and is denoted by  $d_c$ . The behavior of the model is then called *mean-field behavior*. In the self avoiding walk setting we have  $d_c = 4$ , so for  $d > 4$  critical exponents should be equal to the ones of a normal random walk. See [45] for a detailed elaboration on this. Obviously this does not mean that the models are exactly the same. Even though we are in high dimensions, we still can have loops in our random walk. But they are so rare that in the macroscopic picture they are negligible. The intuition carries over for high-dimensional percolation, i.e. the space is so vast or spread out that far away pieces in clusters of our configuration are close to being independent from each other.

In the next part, we want to see what the upper critical dimension  $d_c$  is for percolation on the  $d$ -dimensional lattice and which model has similar behavior/exponents.

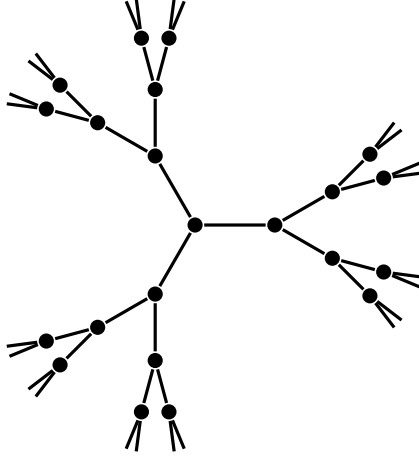


Figure 3.1: Drawing of a subgraph of the 3-regular tree  $\mathbb{T}_3$ , also known as Bethe lattice. Every vertex has exactly 3 neighbors. We see that we can completely split the graph into two disjoint parts just by removing one edge. This is not possible for  $\mathbb{Z}^d$  and is the reason why we can have infinitely many infinite clusters on a regular tree.

For this we present Table 3.1, where we give an overview over some conjectured and partially proven critical exponents. The values for  $d = 2$  were proven for site percolation on the triangular lattice in [46] and for  $d > 6$  we give references later. It is reasonable to believe that Smirnov's results for dimension two also hold in our cubic lattice. From the table we can read that  $d_c = 6$  for our percolation model, since above the values of the exponents do not change anymore. It is typical that the specific values for the exponents are fractions for dimension two and integer values above the upper critical dimension. For the critical dimension we often get the same value but with a logarithmic correction term. Between dimension two and  $d_c$ , the values are often only approached numerically and seem to be irrational.

So how does percolation in high dimension behave? What model has the same exponents? One possibility would be percolation on the  $2d$ -regular tree, denoted by  $\mathbb{T}_{2d}$ . For that we quickly remember that a graph  $G$  is a forest if it contains no cycles and additionally a tree if the graph is also connected (so one single component). Obviously the hypercubic lattice is not loop-free. An example for a regular tree can be found in Figure 3.1. The absence of loops on a tree makes explicit computations possible. Indeed for two vertices  $x$  and  $y$  there exists a unique path connecting them, so if they are  $n$  steps away from each other, we have  $\mathbb{P}_p(x \longleftrightarrow y) = p^n$ . On the other hand, explicitly computing the two point function for percolation on  $\mathbb{Z}^d$  is a true nightmare. Even if we specify the two sites it is

| $d$   | $\beta$        | $\gamma$        | $\delta$       | $\eta$         |
|-------|----------------|-----------------|----------------|----------------|
| 2     | $\frac{5}{36}$ | $\frac{43}{18}$ | $\frac{91}{5}$ | $\frac{5}{24}$ |
| 3 – 6 | ?              | ?               | ?              | ?              |
| $> 6$ | 1              | 1               | 2              | 0              |

Table 3.1: Overview of critical exponents. Question marks refer to a lack of rigorous results.

only possible in special cases.

We can also easily compute the critical exponent  $\gamma$  for the tree. For that we can interpret the cluster of the origin on our percolated tree as a Galton-Watson process. Consider  $\mathbb{T}_r$  with  $r \geq 2$ . The children of the root  $o$  are  $\text{Bin}(r, p)$  distributed, while every other individual's offspring is  $\text{Bin}(r-1, p)$  distributed, denoted by  $\xi$ . We remember, that

- if  $\mathbb{E}[\xi] \leq 1$  our population will die out and
- if  $\mathbb{E}[\xi] > 1$  the population will survive with positive probability.

Using this, the offspring distribution instantly implies that

$$p_c(\mathbb{T}_r) = \frac{1}{r-1}. \quad (3.12)$$

Now let  $|\mathcal{C}_{BP}(o)|$  be the total progeny of a Galton-Watson process with offspring distribution  $\text{Bin}(r-1, p)$  and root  $o$ . Then  $\chi_{BP}(p) := \mathbb{E}[|\mathcal{C}_{BP}(o)|]$  is the susceptibility in this setting. We see that for the cluster size we always get the contribution of the root itself. After that we have  $r-1$  descendants, each with probability  $p$ . This gives us

$$\chi_{BP}(p) = 1 + (r-1)p\chi_{BP}(p), \quad (3.13)$$

from which we can immediately calculate

$$\chi_{BP}(p) = \frac{1}{1 - (r-1)p}. \quad (3.14)$$

If we now come back to the tree setting we have to include the fact that the root has  $r$  possible descendants, yielding

$$\chi(p) = 1 + rp\chi_{BP}(p) = \frac{1+p}{1 - (r-1)p} = \frac{1+p}{r-1}(p_c(\mathbb{T}_r) - p)^{-1}. \quad (3.15)$$

From this we read that  $\gamma = 1$  on  $\mathbb{T}_r$ . Thus the critical exponent  $\gamma$  is the same as the one suggested by Table 3.1. So is percolation on the regular tree the appropriate mean-field model? We claim one can also show that  $\beta$  and  $\delta$  coincide in both models. What about  $\eta$ ? The regular tree has one problem: it lacks a geometric embedding into the euclidean space. This is a bit problematic, since some critical exponents that use euclidean norms (like  $\eta$ ) are then hard to define on a tree.

A better choice would be the *branching random walk* (BRW). For that we follow the construction in [25, Section 2.2] and define a random embedding of percolation on the  $2d$ -regular tree to the euclidean space. To achieve this we want to construct a function  $\phi : \mathbb{T}_{2d} \rightarrow \mathbb{Z}^d$  that maps vertices of the tree into  $\mathbb{Z}^d$ . First we map the root of the tree to the origin, so  $\phi(o) = 0$ . Next for all  $v \in \mathbb{T}_{2d} \setminus \{o\}$  we denote by  $p(v)$  the parent of  $v$ , so the neighbor of  $v$  in the direction to the root. Additionally define a family of i.i.d. random variables  $(Y_v)_{v \in \mathbb{T}_{2d}}$ , where each  $Y_v$  is a random direction/neighbor of the origin, so for every  $y \in \mathbb{Z}^d$  with  $|y| = 1$  we have  $\mathbb{P}(Y_v = y) = 1/2d$ . With this definition we let

$$\phi(v) = \phi(p(v)) + Y_v. \quad (3.16)$$

Then define for  $x \in \mathbb{Z}^d$  the *BRW two-point function*  $G_p(x)$  by

$$G_p(x) = \mathbb{E}_p \left[ \sum_{v \in \mathcal{C}(o)} \mathbb{1}_{\{\phi(v)=x\}} \right], \quad (3.17)$$

where  $\mathbb{E}_p$  is now the expectation with respect to the percolation measure on the tree. With this definition we constructed a percolation configuration in which we ignore cycles, since we only care about the neighborhood structure on the tree. With the same argument as before we can see that also here  $\gamma = 1$ . But now we could also compute critical exponents that use euclidean properties. We claim without a proof that they are indeed the same as on  $\mathbb{Z}^d$ , see [25, Section 2.2].

Time for a bit more rigorous results for percolation in high dimensions. For a function  $f$  and  $n \in \mathbb{N}$  we write  $f^{*n}$  for the  $n$ -fold convolution of  $f$  with itself. It is sometimes useful to express objects we utilize by convolutions, since we can then use the fact that a fourier transform turns convolutions into products. We now introduce the most important convergence condition that we use as a indicator for mean-field behavior. We define the *open triangle diagram* as a convolution by

$$\Delta_p(x) := \tau_p^{*3}(x) = \sum_{z,v \in \mathbb{Z}^d} \mathbb{P}_p(0 \longleftrightarrow z) \mathbb{P}_p(z \longleftrightarrow v) \mathbb{P}_p(v \longleftrightarrow x) \quad (3.18)$$

and the *triangle diagram* by

$$\Delta_p := \Delta_p(0) = \sum_{z,v \in \mathbb{Z}^d} \mathbb{P}_p(0 \longleftrightarrow z) \mathbb{P}_p(z \longleftrightarrow v) \mathbb{P}_p(v \longleftrightarrow 0). \quad (3.19)$$

The term *triangle* comes from the fact that each summand is an upper bound on the probability of forming a triangle. Indeed, to construct a triangle we go from 0 to  $z$ , then from  $z$  to  $v$  and finally back to 0, using disjoint edges. The BK inequality then yields

$$\begin{aligned} \mathbb{P}_p(\text{triangle between } 0, z \text{ and } v) &= \mathbb{P}_p(\{0 \longleftrightarrow z\} \circ \{z \longleftrightarrow v\} \circ \{v \longleftrightarrow 0\}) \\ &\leq \mathbb{P}_p(0 \longleftrightarrow z) \mathbb{P}_p(z \longleftrightarrow v) \mathbb{P}_p(v \longleftrightarrow 0), \end{aligned} \quad (3.20)$$

so the triangle diagram is an upper bound on the probability of all triangles having the origin as a corner. Of course these are not triangles in the classical sense with straight lines. Think of triangles as loops with three fixed points. The *triangle condition* we want to use as a sign for mean-field behavior is now the finiteness of the triangle diagram at criticality, so  $\Delta_{p_c} < \infty$ . This was first introduced in [3] and we follow that the probability of circles in form of triangles should be reasonably low. Note this implies that

$$\tau_{p_c}(x) = 0, \quad \text{as } |x| \rightarrow \infty, \quad (3.21)$$

which in turn means  $\theta(p_c) = 0$ , since the probability of all long range connections vanish. The next lemma gives an upper bound on the critical triangle in high dimensions.

**Lemma 3.2** (Bound on  $\Delta_p$  [21]). *For percolation there exist  $d_0 > 6$  and  $C_{p_c} > 0$  such that for  $d \geq d_0 > 6$*

$$\Delta_{p_c} \leq 1 + \frac{C_{p_c}}{d}. \quad (3.22)$$

It is widely believed that the triangle condition holds if and only if  $d > 6$ . A lot of results heavily point in that direction and the term “true conjecture” would be fitting. The first such threshold, above the triangle condition is fulfilled was obtained by Hara and Slade in [21]. The original paper just states that the condition holds if we choose  $d$  large enough. The explicit dimension was just stated indirectly, but experts working in the field knew that  $d \geq 19$  would suffice. For a long period no further improvements were made. Currently, the best threshold was shown more than 20 years later by Fitzner and van der Hofstad [15].

**Theorem 3.3** (Finite triangle [15]). *For  $d > 10$  the triangle condition holds.*

The proof relies on a method called *lace expansion*. It is a powerful tool also used for self avoiding random walks, but we will not go into detail here. For a more in-depth treatment we can again refer to [25] and for more on the lace expansion including self avoiding walks we recommend [45]. The triangle is also finite for long range percolation with  $d > 6$  and  $L$  being large enough, also shown in [21], which is the reason a lot of the presented results also hold in this case.

Normally we are not interested in the exact value of the upper and lower constants occurring when talking about critical exponents in the bounded ratio sense. Most of the time it suffices to show their existence, so we then have bounds that we can use. We will later see a case where the behavior of the constants with respect to  $d$  will matter for  $\gamma$ . This is solved by the next theorem.

**Theorem 3.4** ( $\gamma = 1$  given the triangle condition [3]). *If the triangle condition holds, the critical exponent  $\gamma$  exists in the bounded-ratio sense and takes on the mean-field value  $\gamma = 1$ . If we further assume that the condition holds in the form that for  $e \sim 0$  the open triangle diagram satisfies  $\Delta_{p_c}(e) < 1$ , we have for all  $p \in [0, p_c)$*

$$\frac{1}{2d}(p_c - p)^{-1} \leq \chi(p) \leq \frac{1}{2d(1 - \Delta_{p_c}(e))}(p_c - p)^{-1}. \quad (3.23)$$

Note that from (3.23) we deduce  $\gamma = 1$ , but also the exact constants. The specific constants for the other exponents introduced will not be important for us. Thus we summarize them in the next theorem.

**Theorem 3.5** ( $\delta = 2$ ,  $\beta = 1$  and  $\eta = 0$  [3, 6, 20]). *If the triangle condition holds, the critical exponents  $\delta$ ,  $\beta$  and  $\eta$  exist in the bounded-ratio sense and take on their mean-field values  $\delta = 2$ ,  $\beta = 1$  and  $\eta = 0$ .*

The statement for  $\eta$  was shown in an even stronger way. Hara proved the right asymptotic behavior in [20].

To summarize, we can conclude that the behavior of our model does not only change drastically around  $p_c$ , but also when we have mean-field behavior above  $d_c$ . We saw that under the triangle condition we have rigorous results for our critical exponents. Even though it is not obvious at first, using the triangle condition as an indicator for mean-field behavior works quite well. With all these useful upper and lower bounds on important objects in high dimensions we are well prepared for the next chapter, where they are applied.

# Chapter 4

## The One-Arm Exponent

We now study the main object of this thesis. Starting from a point we are interested in the probability of the existence of a shortest path with a certain length. Thus we want to analyze the behavior of so called *arm events*. For  $x, y \in \mathbb{Z}^d$  we define  $d_{\text{int}}(x, y)$  to be the *intrinsic distance* between  $x$  and  $y$ . That is the length of the shortest path connecting  $x$  and  $y$  on the percolated subgraph. We should be aware that this path does not have to be unique. If no such path exists, set  $d_{\text{int}}(x, y) = \infty$ . Many other names for this notion are used in the literature: shortest path metric, graph distance, hop-count distance, geodesic<sup>1</sup> length and the physics literature introduced the term chemical distance. Note that this distance depends on the given percolation configuration and is therefore random and not a deterministic metric on  $\mathbb{Z}^d$ . Therefore the name intrinsic distance. Additionally, we define the *intrinsic radius* of the cluster of  $x$  by

$$\text{rad}_{\text{int}}(x) = \sup_{y \in \mathcal{C}(x)} d_{\text{int}}(x, y). \quad (4.1)$$

A bit informally one could say we are interested in the longest shortest path. This path is then called an *arm*. Sometimes the literature uses the cluster of  $x$  as an argument, so the notation  $\text{rad}_{\text{int}}(\mathcal{C}(x))$  is used. This makes sense since the radius is with respect to the cluster and  $x$  as its central point. But considering that  $\mathcal{C}(x) = \mathcal{C}(y)$  can hold but  $\text{rad}_{\text{int}}(\mathcal{C}(x)) \neq \text{rad}_{\text{int}}(\mathcal{C}(y))$ , we thought this is maybe a bit inconsistent or ill-defined.

In this chapter we want to study how the probability of intrinsic radii of certain length behave with respect to the desired length. The corresponding events are then given by:

**Definition.** *The intrinsic one-arm event for  $n \in \mathbb{N}$  is given by the set  $\{\text{rad}_{\text{int}}(x) \geq n\}$ .*

If we are only interested in an arbitrary vertex, we use the shorter notation  $\text{rad}_{\text{int}} \equiv \text{rad}_{\text{int}}(0)$  and  $d_{\text{int}}(x) \equiv d_{\text{int}}(0, x)$ . The intrinsic arm event might look innocent at first, but it can be complicated. It is obvious that the event is not decreasing. Indeed, if we make edges vacant we could destroy a path which we need for our arm event to hold. One might think at first that the events are increasing. Surely making more edges occupied makes the occurrence of paths more likely? This is true, but the problem is that we measure the shortest path and we could destroy a very long path if we create a shortcut and build a faster way to get to that site. This is visualized in Figure 4.1. We follow that the events are neither increasing or decreasing and we cannot use any machinery tailored for these

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<sup>1</sup>A geodesic is the shortest curve or path connecting two points.

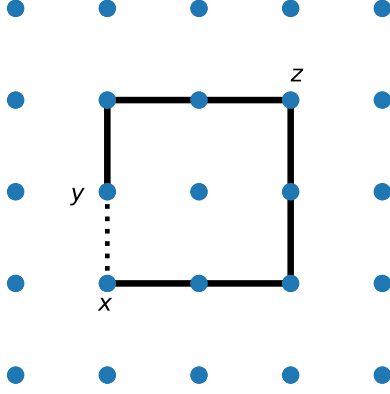


Figure 4.1: Since the intrinsic distance between  $x$  and  $y$  is 7, we have  $\text{rad}_{\text{int}}(x) = 7$ . If the dotted line is added to the graph we get  $d_{\text{int}}(x, y) = 1$  and the longest arm is now between  $x$  and  $z$ , so  $\text{rad}_{\text{int}}(x) = 4$ . This creation of shorter paths is prohibiting our events from being increasing.

settings. In a related fashion it is unknown, whether for a fixed  $n$  the quantity  $\mathbb{P}_p(x \xleftrightarrow{n} y)$  is monotone in  $p$ . Here  $\{x \xleftrightarrow{n} y\}$  refers to the events where the shortest path between  $x$  and  $y$  has precisely length  $n$ . The problem remains the same: if we increase  $p$  there could arise new connections of length  $n$ , but on the other hand a faster way of getting from  $x$  to  $y$  could appear.

The main question now is, how the probability of such an event depends on the length  $n$  of our path/arm. Surely the occurrence of an arm event has a non-trivial probability. Obviously the event is more likely than a straight line of length  $n$  in an arbitrary direction, since straight lines are always geodesic in our setting. On the other hand the worst case would be if  $|\mathcal{C}(0)| = 1$ , i.e. the origin is completely isolated, thus

$$0 < p_c^n \leq \mathbb{P}_{p_c}(\text{rad}_{\text{int}} \geq n) \leq 1 - (1 - p_c)^{2d} < 1. \quad (4.2)$$

Since  $\{\text{rad}_{\text{int}} \geq n\} \subset \{\text{rad}_{\text{int}} \geq n - 1\}$ , the probability is certainly decreasing in  $n$ , but we would like to have a more quantitative expression for this. How fast is the probability decreasing in  $n$ ? For this we need a new critical exponent, as we promised earlier.

**Definition.** Define the intrinsic one-arm critical exponent  $\rho_{\text{in}} > 0$  as

$$\mathbb{P}_{p_c}(\text{rad}_{\text{int}} \geq n) \asymp n^{-1/\rho_{\text{in}}}, \text{ as } n \rightarrow \infty. \quad (4.3)$$

## 4.1 Arm Exponent Mean-Field Value

Before we focus on  $\mathbb{Z}^d$ , we could already guess a value for  $\rho_{\text{in}}$  by first looking at the corresponding exponent for the regular tree as our mean-field model. Then we will predict that the value is the same on the hypercubic lattice. The definition naturally transfers over, because it is a property of the graph and does not need the euclidean embedding. This classical result was first shown by Kolmogorov in 1938, however we present a different proof.

**Proposition 4.1** ( $\rho_{\text{in}} = 1$  on a tree [32]). *On  $\mathbb{T}_r$  with  $r \geq 3$  we have*

$$\lim_{n \rightarrow \infty} n \mathbb{P}_{p_c}(\text{rad}_{\text{int}} \geq n) = \frac{2r - 2}{r - 2}. \quad (4.4)$$

*Proof.* For  $n \geq 1$  define

$$\theta_n = \mathbb{P}_{p_c}(\text{rad}_{\text{int}} \geq n) = \mathbb{P}_{p_c}(\exists v \in \mathcal{C}_{BP}(o) \text{ such that } h(v) = n), \quad (4.5)$$

where  $h(\cdot)$  is the distance to the root. We can get a recursive representation of  $\theta_n$  if we write

$$\begin{aligned} 1 - \theta_n &= \mathbb{P}_{p_c}(\nexists v \in \mathcal{C}_{BP}(o) \text{ such that } h(v) = n) \\ &= \mathbb{P}_{p_c}(\text{we do not reach generation } n) \\ &= \mathbb{P}_{p_c}(\text{we do not reach generation } n-1 \text{ and one subsequent edge is open}) \\ &= (1 - p_c \theta_{n-1})^{r-1}. \end{aligned} \quad (4.6)$$

It is left so show that this recursion implies the asymptotic statement in (4.4). For that define

$$f(x) = 1 - (1 - p_c x)^{r-1}, \quad (4.7)$$

what can be used to recreate the recursive form

$$\theta_n = f(\theta_{n-1}). \quad (4.8)$$

Since  $f'(x) = (1 - p_c x)^{r-2} < 1$  for  $x \in (0, 1]$ , we have  $f(x) < x$  for all  $x \in (0, 1]$ . This implies that  $\theta_n$  is decreasing to 0. If we now use Taylor's formula on  $f$  we get

$$f(x) = x - (b + o(1))x^2, \quad \text{as } x \rightarrow 0, \quad (4.9)$$

for  $b = (r-2)/(2r-2)$ . Hence for all  $n$  we can find  $a_n$  such that  $a_n \rightarrow b$  when  $n \rightarrow \infty$  and

$$\theta_n = \theta_{n-1} - a_n \theta_{n-1}^2. \quad (4.10)$$

Setting  $c_n = n\theta_n$  for  $n \in \mathbb{N}$  and using (4.10) we can find  $b_n$  with  $b_n \rightarrow b$  when  $n \rightarrow \infty$  such that

$$c_n = \frac{n}{n-1} c_{n-1} - \frac{b_n}{n} c_{n-1}^2. \quad (4.11)$$

Since  $c_n/n = \theta_n$  decreases to 0 we can follow from (4.11) that

$$\frac{c_n}{c_{n-1}} \xrightarrow{n \rightarrow \infty} 1 \quad (4.12)$$

has to be fulfilled, so we can exclude oscillating behavior. Let  $h \in (0, 1)$ . We will now show that the sequence  $(c_{n-1})_{n \in \mathbb{N}}$  is confined (at least the accumulation points) in between the left moving barrier  $(c_n^l)_{n \in \mathbb{N}}$  and the right moving barrier  $(c_n^r)_{n \in \mathbb{N}}$  given by

$$c_n^l := \frac{n}{n-1} \frac{1-h}{b_n}, \quad c_n^r := \frac{n}{n-1} \frac{1+h}{b_n}. \quad (4.13)$$

Assume there is  $n \geq 3$ , such that  $c_{n-1} \leq c_n^l$  holds. Using (4.11) we can then estimate

$$\frac{c_n}{c_{n-1}} \geq 1 + \frac{h}{n-1} > 1. \quad (4.14)$$

Assume now that  $c_{n-1} \leq c_n^l$  holds for all consecutive  $n \in \mathbb{N}$ . Then we can write  $c_n$  as a telescopic product and use the fact that  $\prod_{j=n}^{\infty} (1 + \frac{h}{j-1}) = \infty$  to get that  $c_n$  diverges. But this would be contradictory to the convergence of the upper bound on  $c_n$ :

$$c_n^l \xrightarrow{n \rightarrow \infty} \frac{1-h}{b} < \infty. \quad (4.15)$$

Thus there will be some natural number  $m \geq n$  such that

$$c_{n-1} \leq c_n^l, \dots, c_{m-1} \leq c_m^l, c_m > c_{m+1}^l \text{ and } c_{n-1} < c_n < \dots < c_m. \quad (4.16)$$

Informally, if we exceed the left barrier, we return monotonically in finite time. We can now do the same procedure for the upper barrier. Assume there is  $n \geq 3$  such that  $c_{n-1} \geq c_n^r$  holds. Using (4.11) we can then again estimate

$$\frac{c_n}{c_{n-1}} \leq 1 - \frac{h}{n-1} < 1. \quad (4.17)$$

If  $c_{n-1} \geq c_n^r$  holds for all consecutive  $n \in \mathbb{N}$ , we can write  $c_n$  as a telescopic product and use that  $\prod_{j=n}^{\infty} (1 - \frac{h}{j-1}) = 0$  to get that  $c_n$  converges to 0. But this would be contradictory to the convergence of the lower bound on  $c_n$ :

$$c_n^r \xrightarrow{n \rightarrow \infty} \frac{1+h}{b} > 0. \quad (4.18)$$

Thus there will be some natural number  $s \geq n$  such that

$$c_{n-1} \geq c_n^r, \dots, c_{s-1} \geq c_s^r, c_s < c_{s+1}^r \text{ and } c_{n-1} > c_n > \dots > c_s. \quad (4.19)$$

Informally, if we exceed the right barrier, we return monotonically in finite time. We can sum this up by concluding that for any  $h \in (0, 1)$

$$\frac{1-h}{b} \leq \liminf_n c_n \leq \limsup_n c_n \leq \frac{1+h}{b}. \quad (4.20)$$

This instantly shows that  $c_n$  converges to  $1/b$  and by the definition of  $c_n$  we see that  $\theta_n \sim 1/(nb)$ , finishing the proof.  $\square$

Note that the leading order term of (4.4) also implies that  $\rho_{\text{in}} = 1$  in the bounded-ratio sense on a tree. The shown statement is even stronger, since we know the correct asymptotic constant. Thus, mean-field behavior should imply that  $\rho_{\text{in}} = 1$  is the correct value, when we look at the hypercubic lattice for  $d > 6$ . This is the content of the next theorem.

**Theorem 4.2** ( $\rho_{\text{in}} = 1$  under the triangle condition [35]). *If the triangle condition holds, there exist constants  $0 < c_{\text{in}} < C_{\text{in}} < \infty$  such that for all  $n \in \mathbb{N}$*

$$\frac{c_{\text{in}}}{n} \leq \mathbb{P}_{p_c}(\text{rad}_{\text{int}} \geq n) \leq \frac{C_{\text{in}}}{n}. \quad (4.21)$$

Hence the critical exponent  $\rho_{\text{in}}$  exists in the bounded-ratio sense and takes on the mean-field value  $\rho_{\text{in}} = 1$ .

The statement was proven not too long ago in 2009 in the breakthrough work of Kozma and Nachmias [35] and from this they were able to derive the so called *Alexander-Orbach conjecture*<sup>2</sup>, which was introduced by Alexander and Orbach in [4]. For this they first

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<sup>2</sup>They showed that the *spectral dimension*  $d_s$  on the incipient infinite cluster (IIC) for high dimensions is equal to  $4/3$ . The mean-field analogue was shown in [5]. The spectral dimension of an infinite connected Graph  $G$  is defined by

$$d_s = d_s(G) = -2 \lim_{n \rightarrow \infty} \frac{\log p_{2n}(x, x)}{\log n}, \quad (4.22)$$

where  $x \in G$  and we write for  $p_n(x, x)$  for the return probability of the simple random walk in  $G$  after  $n$  steps. For the IIC one examines the weak limit of conditional percolation measures: For critical percolation, condition on the event that  $0 \longleftrightarrow x$  and let  $|x| \rightarrow \infty$ . For more details and further references see the introduction in [35].

established bounds on the expected size of *intrinsic balls*. Define the intrinsic ball around  $x \in \mathbb{Z}^d$  with radius  $n$  by

$$B(x, n) := \{y \in \mathbb{Z}^d \mid d_{\text{int}}(x, y) \leq n\}. \quad (4.23)$$

By translation invariance, the exact center of the ball is often irrelevant. In this case we can (as done before) use 0 as our reference point and write  $B(n) = B(0, n)$ . These balls are in fact monotone in the sense that if  $x \in B(n)$  we can add arbitrary edges to the graph and  $x \in B(n)$  will still hold. With this definition we can interpret the arm event as the probability that we have a connection linking the boundary of the intrinsic ball and the origin. This is because

$$\mathbb{P}_{p_c}(\text{rad}_{\text{int}} \geq n) = \mathbb{P}_{p_c}(0 \longleftrightarrow \partial B(n)), \quad (4.24)$$

where  $\partial$  refers to the vertex boundary of  $B(n)$  defined by

$$\partial B(n) = B(n) \setminus B(n-1) = \{y \in \mathbb{Z}^d \mid d_{\text{int}}(0, y) = n\}. \quad (4.25)$$

Thus, a good grasp of the intrinsic balls can help us analyzing arm events. In [26] the second author claims this was first introduced in [39] to study the mixing time of a lazy simple random walk on the largest component of a critical Erdős-Rényi graph. There the subsequent upper and lower bounds were used as assumptions and were later confirmed through proofs that we will present.

**Theorem 4.3** (Intrinsic expected volume [24, 35, 43]). *If the triangle condition holds, there exist constants  $0 < c_b < C_b < \infty$  such that for all  $n \in \mathbb{N}$*

$$c_b n \leq \mathbb{E}_{p_c}|B(n)| \leq C_b n. \quad (4.26)$$

Since

$$|B(n)| = \sum_{k=0}^n |\partial B(k)| \quad (4.27)$$

and because of the upper bound of (4.26) it is believed in [27] that the sequence  $\mathbb{E}_{p_c}|\partial B(k)|$  is bounded. However it is an open problem to show that we can follow this from the triangle condition.

Sapozhnikov found a really elegant way to derive the upper bound of (4.26) using a monotone coupling [43] and we will show his strategy for that part. Not only is the proof significantly shorter, but also the derived constant  $C_b$  sharper than the one by Kozma and Nachmias.

*Proof of Theorem 4.3.* We use a coupling of critical percolation and subcritical  $p < p_c$ . We start at percolation for  $p_c$  and write  $\omega_{p_c}$  for the corresponding percolation configuration. We construct the configuration  $\omega_p$  as follows: Make every occupied edge of  $\omega_{p_c}$  vacant with probability  $1 - (p/p_c)$ . This guaranties the correct marginal distribution. We write  $\mathbb{P}_{(p_c, p)}$  for the joint distribution and use superscripts to label the configuration on which the events appear, i.e. for  $\{d_{\text{int}}(x) \geq n\}$  on  $\omega_p$ , write  $\{d_{\text{int}}^{\omega_p}(x) \geq n\}$ . With this we get for all  $x \in \mathbb{Z}^d$  and  $n \in \mathbb{N}$

$$\begin{aligned} \mathbb{P}_p(d_{\text{int}}(x) \leq n) &= \mathbb{P}_{(p_c, p)}(d_{\text{int}}^{\omega_p}(x) \leq n) \\ &= \mathbb{P}_{(p_c, p)}(d_{\text{int}}^{\omega_p}(x) \leq n \mid d_{\text{int}}^{\omega_{p_c}}(x) \leq n) \mathbb{P}_{p_c}(d_{\text{int}}(x) \leq n). \end{aligned} \quad (4.28)$$

If we have a shortest path of length  $n$  connecting  $0$  and  $x$  in  $\omega_{p_c}$ , the exact same path remains open with probability  $(p/p_c)^n$  when switching to  $\omega_p$ . Thus

$$\mathbb{P}_p(d_{\text{int}}(x) \leq n) \geq \left(\frac{p}{p_c}\right)^n \mathbb{P}_{p_c}(d_{\text{int}}(x) \leq n). \quad (4.29)$$

Summing over  $x$  and using the inequality  $\mathbb{P}_p(d_{\text{int}}(x) \leq n) \leq \mathbb{P}_p(0 \longleftrightarrow x)$  yields

$$\mathbb{E}_{p_c}|B(n)| = \sum_x \mathbb{P}_{p_c}(d_{\text{int}}(x) \leq n) \leq \left(\frac{p_c}{p}\right)^n \chi(p) \leq C_\gamma \left(\frac{p_c}{p}\right)^n (p_c - p)^{-1}, \quad (4.30)$$

where we used that  $\gamma = 1$  from Theorem 3.4. We see that we can take  $C_b = 4C_\gamma/p_c$ , if we choose  $p = p_c(1 - \frac{1}{2n})$ .

The next part now follows Kozma and Nachmias original proof. We first estimate the conditional expectation  $\mathbb{E}_{p_c}[d_{\text{int}}(x) | 0 \longleftrightarrow x]$ . If  $\{0 \longleftrightarrow x\}$  occurs, then  $d_{\text{int}}(x)$  is bounded from above by the number of  $y \in \mathbb{Z}^d$  such that the events  $\{0 \longleftrightarrow y\}$  and  $\{y \longleftrightarrow x\}$  occur disjointly. Using the BK inequality and  $\eta = 0$  yields

$$\mathbb{E}_{p_c}[d_{\text{int}}(x) \mathbb{1}(0 \longleftrightarrow x)] \leq \sum_{y \in \mathbb{Z}^d} \mathbb{P}_{p_c}(\{0 \longleftrightarrow y\} \circ \{y \longleftrightarrow x\}) \leq C_\gamma^2 \sum_{y \in \mathbb{Z}^d} |y|^{2-d} |x - y|^{2-d}. \quad (4.31)$$

With the definition of  $|\cdot|$  and the Hölder inequality we can perform a straightforward calculation and bound the right hand side of (4.31) by  $C|x|^{4-d}$  for some constant  $C > 0$ . Using  $\eta = 0$  another time, we compute

$$\mathbb{E}_{p_c}[d_{\text{int}}(x) | 0 \longleftrightarrow x] = \frac{\mathbb{E}_{p_c}[d_{\text{int}}(x) \mathbb{1}(0 \longleftrightarrow x)]}{\mathbb{P}_{p_c}(0 \longleftrightarrow x)} \leq C|x|^2. \quad (4.32)$$

We use this and the conditional version of Markov's inequality to conclude that for  $|x| \leq \sqrt{n/2C}$  we have

$$\begin{aligned} \mathbb{P}_{p_c}(d_{\text{int}}(x) \leq n | 0 \longleftrightarrow x) &= 1 - \mathbb{P}_{p_c}(d_{\text{int}}(x) > n | 0 \longleftrightarrow x) \\ &\geq 1 - \frac{\mathbb{E}_{p_c}[d_{\text{int}}(x) | 0 \longleftrightarrow x]}{n} \\ &\geq \frac{1}{2}. \end{aligned} \quad (4.33)$$

Thus we can estimate

$$\mathbb{E}_{p_c}|B(n)| \geq \sum_{x: |x| \leq \sqrt{n/2C}} \mathbb{P}_{p_c}(d_{\text{int}}(x) \leq n \text{ and } 0 \longleftrightarrow x) \geq \frac{c_\eta}{2} \sum_{x: |x| \leq \sqrt{n/2C}} |x|^{2-d}. \quad (4.34)$$

Next we can use the well known fact that  $\|x\|_\infty \leq \|x\|_2 \leq \sqrt{d}\|x\|_\infty$  and count points inside a ball with respect to the  $\ell^\infty$ -norm to conclude

$$\sum_{x: |x| \leq \sqrt{n/2C}} |x|^{2-d} \geq \sum_{x: \sqrt{d}\|x\|_\infty \leq \sqrt{n/2C}} \|x\|_\infty^{d-2} \geq c'n, \quad (4.35)$$

for some constant  $c' > 0$ , finishing the proof.  $\square$

We can refer to [24, Theorem 2.1] for an alternative proof. There  $c_b = 1$  is shown and the bound is more general because it holds for all  $d \geq 1$ .

Even though Theorem 4.3 can be used for the one-arm estimate, we are able to use the coupling in a more direct way to prove it even without any intrinsic ball growth statement. For this we got inspired by Sapozhniko [43] and Hutchcroft [29], who used a similar coupling. Since we will use the upper bound later on, we did not skip the proof of Theorem 4.3 in sake of completeness.

*Lower bound of Theorem 4.2 using coupling.* Let  $p = p_c + 1/n$ . First note that from Theorem 3.5 we know that under the triangle condition we have  $\beta = 1$ , thus

$$\mathbb{P}_p(\text{rad}_{\text{int}} \geq n) \geq \mathbb{P}_p(|\mathcal{C}(0)| = \infty) = \theta(p) \geq \frac{c_\beta}{n}. \quad (4.36)$$

This gives us our lower bound for  $p$ . To get one for the correct probability  $p_c$ , we use a similar coupling as in the proof of Theorem 4.3, but here we couple critical percolation with supercritical. Starting from percolation at  $p = p_c + 1/n$ , remove occupied edges with probability  $1 - (p_c/p)$ . Let us assume that  $\{\text{rad}_{\text{int}} \geq n\}$  occurs for the configuration with parameter  $p$ , then there exists a path of minimal length  $n$  connecting 0 and  $\partial B(n)$ . With probability  $(p_c/p)^n$  the exact same path is still active when switching to the critical percolation configuration. Thus, as in (4.28)

$$\mathbb{P}_{p_c}(\text{rad}_{\text{int}} \geq n) \geq \left(\frac{p_c}{p}\right)^n \mathbb{P}_p(\text{rad}_{\text{int}} \geq n). \quad (4.37)$$

Because

$$\left(\frac{p_c}{p}\right)^n = \left(\frac{p_c}{p_c + \frac{1}{n}}\right)^n \xrightarrow{n \rightarrow \infty} e^{-\frac{1}{p_c}}, \quad (4.38)$$

we can choose  $c_{\text{in}} > 0$  independently of  $n$ , which was what we wanted.  $\square$

We now show a different way to obtain a lower bound, using the very useful *second moment method* together with Theorem 4.3.

*Alternative proof of the lower bound of Theorem 4.2.* First we remember the inequality used by the second moment method. For that let  $X$  be an almost surely positive random variable. Then we can use Cauchy-Schwarz and get

$$\mathbb{E}X = \mathbb{E}[X \mathbb{1}(X > 0)] \leq (\mathbb{E}[X^2])^{1/2} \mathbb{P}(X > 0)^{1/2}. \quad (4.39)$$

Rewriting gives us the useful expression (also known as Paley–Zygmund inequality)

$$\mathbb{P}(X > 0) \geq \frac{(\mathbb{E}X)^2}{\mathbb{E}X^2}. \quad (4.40)$$

Picking  $X = |B(\lambda n) \setminus B(n)|$  with  $\lambda = 2C_b/c_b$  we can use Theorem 4.3 and receive

$$\mathbb{E}_{p_c}|B(\lambda n)| \geq c_b \lambda n \quad \text{and} \quad \mathbb{E}_{p_c}|B(n)| \leq C_b n. \quad (4.41)$$

Combining these estimates yields

$$\mathbb{E}_{p_c}|B(\lambda n) \setminus B(n)| \geq c_b \lambda n - C_b n = C_b n. \quad (4.42)$$

Note that

$$\begin{aligned} & \{d_{\text{int}}(0, x) \leq \lambda n\} \cap \{d_{\text{int}}(0, y) \leq \lambda n\} \\ & \subseteq \bigcup_{z \in \mathbb{Z}^d} \{d_{\text{int}}(0, z) \leq \lambda n\} \circ \{d_{\text{int}}(z, x) \leq \lambda n\} \circ \{d_{\text{int}}(z, y) \leq \lambda n\}. \end{aligned} \quad (4.43)$$

Using this and the BK inequality (2.8), we estimate

$$\begin{aligned} \mathbb{E}_{p_c} |B(\lambda n)|^2 &= \sum_{x, y} \mathbb{P}_{p_c}(d_{\text{int}}(0, x) \leq \lambda n, d_{\text{int}}(0, y) \leq \lambda n) \\ &\leq \sum_{x, y, z} \mathbb{P}_{p_c}(d_{\text{int}}(0, z) \leq \lambda n) \mathbb{P}_{p_c}(d_{\text{int}}(z, x) \leq \lambda n) \mathbb{P}_{p_c}(d_{\text{int}}(z, y) \leq \lambda n) \\ &= \sum_{x, y, z} \mathbb{P}_{p_c}(d_{\text{int}}(0, z) \leq \lambda n) \mathbb{P}_{p_c}(d_{\text{int}}(0, x - z) \leq \lambda n) \mathbb{P}_{p_c}(d_{\text{int}}(0, y - z) \leq \lambda n) \\ &= \sum_{x, y, z} \mathbb{P}_{p_c}(d_{\text{int}}(0, z) \leq \lambda n) \mathbb{P}_{p_c}(d_{\text{int}}(0, x) \leq \lambda n) \mathbb{P}_{p_c}(d_{\text{int}}(0, y) \leq \lambda n) \\ &= \left[ \sum_{z \in \mathbb{Z}^d} \mathbb{P}_{p_c}(d_{\text{int}}(0, z) \leq \lambda n) \right]^3 = \left[ \mathbb{E}_{p_c} |B(\lambda n)| \right]^3 \leq C' n^3, \end{aligned} \quad (4.44)$$

for some constant  $C' > 0$ . If we write

$$\mathbb{P}_{p_c}(\text{rad}_{\text{int}} \geq n) \geq \mathbb{P}_{p_c}(|B(\lambda n) \setminus B(n)| > 0) \stackrel{(4.40)}{\geq} \frac{(\mathbb{E}_{p_c} |B(\lambda n) \setminus B(n)|)^2}{\mathbb{E}_{p_c} |B(\lambda n) \setminus B(n)|^2}, \quad (4.45)$$

we can see that using  $\mathbb{E}_{p_c} |B(\lambda n) \setminus B(n)| \leq \mathbb{E}_{p_c} |B(\lambda n)|$  with (4.42) and (4.44) yields

$$\mathbb{P}_{p_c}(\text{rad}_{\text{int}} \geq n) \geq \frac{C_{\text{in}}^2 n^2}{C' n^3} \geq \frac{c_{\text{in}}}{n}, \quad (4.46)$$

for some constant  $c_{\text{in}} > 0$ . □

One big obstacle in [35] was the derivation of the upper bound of (4.21). Actually, for the Alexander-Orbach conjecture only the upper bounds of (4.21) and (4.26) were needed and the lower ones were just stated for the sake of completeness. Remember that our intrinsic arm events were not monotone in the sense that we can create shorter paths? To overcome this problem, we define for subgraphs  $G \subseteq \mathcal{E}(\mathbb{Z}^d)$

$$H(n; G) := \{0 \longleftrightarrow \partial B(n; G)\} \quad (4.47)$$

for the one-arm event on the graph  $G$ , where  $B(n; G)$  is the intrinsic ball around 0 with radius  $n$  in  $G$ . Further define

$$\Gamma(n) := \sup_{G \subseteq \mathcal{E}(\mathbb{Z}^d)} \mathbb{P}_{p_c}(H(n; G)). \quad (4.48)$$

We note that even though we are looking at all subgraphs of  $\mathbb{Z}^d$ , we do not use the graph specific critical value  $p_c(G)$ . Instead we use  $p_c(\mathbb{Z}^d)$  in all cases. We work with  $\Gamma(n)$  instead of  $\mathbb{P}_{p_c}(0 \longleftrightarrow \partial B(n))$ , since we will condition on the specific  $B(j, G)$  and need a uniform upper bound holding for all subgraphs. This monotonizing strategy was suggested by Yuval Peres.

Upper bound of Theorem 4.2. Choose  $A \geq 1$  large enough such that

$$3^3 A^{2/3} + C_\delta A^{2/3} \leq A, \quad (4.49)$$

where  $C_\delta$  is the upper bound from  $\delta = 2$  (it holds that  $C_\delta = \sqrt{12}e/(e-1)$ , so  $A \geq 34266$  would suffice). We will show that for every integer  $k \geq 0$

$$\Gamma(3^k) \leq \frac{A}{3^k}. \quad (4.50)$$

This inequality actually would finish our proof, since for  $n \in \mathbb{N}$  we can now choose  $3^{k-1} \leq n \leq 3^k$  and receive

$$\mathbb{P}_{p_c}(\text{rad}_{\text{int}} \geq n) = \mathbb{P}_{p_c}(H(n; \mathbb{Z}^d)) \leq \Gamma(n) \leq \Gamma(3^{k-1}) \leq \frac{A}{3^{k-1}} \leq \frac{3A}{n}, \quad (4.51)$$

completing the proof with  $C_{\text{in}} = 3A$ . We show (4.50) by induction. For  $k = 0$  there is nothing to prove, since we bound a probability from above by 1. So we now assume that the claim holds for  $k-1$  and we will show that it also holds for  $k$ . Let  $G \subset \mathcal{E}(\mathbb{Z}^d)$  be any subgraph and write  $|\mathcal{C}_G(0)|$  for the size of the cluster of 0 in  $G$ . This yields

$$\begin{aligned} \mathbb{P}_{p_c}(H(3^k; G)) &= \mathbb{P}_{p_c}(H(3^k; G), |\mathcal{C}_G(0)| \leq A^{-4/3}9^k) + \mathbb{P}_{p_c}(H(3^k; G), |\mathcal{C}_G(0)| > A^{-4/3}9^k) \\ &\leq \mathbb{P}_{p_c}(H(3^k; G), |\mathcal{C}_G(0)| \leq A^{-4/3}9^k) + \mathbb{P}_{p_c}(|\mathcal{C}_G(0)| > A^{-4/3}9^k). \end{aligned} \quad (4.52)$$

We can bound the second summand if we remember from Theorem 3.5 that  $\delta = 2$  under the triangle condition:

$$\mathbb{P}_{p_c}(|\mathcal{C}_G(0)| > A^{-4/3}9^k) \leq \mathbb{P}_{p_c}(|\mathcal{C}(0)| \geq A^{-4/3}9^k) \leq C_\delta A^{2/3}3^{-k}. \quad (4.53)$$

Note that if  $|\mathcal{C}_G(0)| \leq A^{-4/3}9^k$ , there must be some level  $j \in [\frac{1}{3}3^k, \frac{2}{3}3^k]$  such that  $|\partial B(j; G)| \leq A^{-4/3}3^{k+1}$ . Otherwise we would observe too many points for  $\mathcal{C}_G(0)$ . Denote the first such (random but unique) level by  $j$ . Then we can condition and sum over all possible values of  $B(j; G)$  and get

$$\begin{aligned} \mathbb{P}_{p_c}(H(3^k; G), |\mathcal{C}_G(0)| \leq A^{-4/3}9^k) &= \\ \sum_{\mathcal{A} \subset \mathcal{E}(\mathbb{Z}^d)} \mathbb{P}_{p_c}(H(3^k; G), |\mathcal{C}_G(0)| \leq A^{-4/3}9^k \mid B(j; G) = \mathcal{A}) \mathbb{P}_{p_c}(B(j; G) = \mathcal{A}). \end{aligned} \quad (4.54)$$

Note that if  $\partial B(3^k; G) \neq \emptyset$ , then at least one vertex  $v$  at level  $j$  has to reach level  $3^{k-1}$ , i.e. the path of our arm event still has length  $3^{k-1}$  left to go. If we then define  $G_2$  to be  $G$  with all edges needed to calculate  $B(j; G)$  removed and use the definition of  $j$  we deduce

$$\begin{aligned} &\mathbb{P}_{p_c}(H(3^k; G), |\mathcal{C}_G(0)| \leq A^{-4/3}9^k \mid B(j; G)) \\ &\leq \mathbb{P}_{p_c}\left(\bigcup_{v \in \partial B(j; G)} \{\partial B(v, 3^{k-1}; G_2) \neq \emptyset\} \mid B(j; G)\right) \\ &\leq A^{-4/3}3^{k+1} \mathbb{P}_{p_c}(\partial B(0, 3^{k-1}; G_2) \neq \emptyset) \\ &\leq A^{-4/3}3^{k+1} \Gamma(3^{k-1}). \end{aligned} \quad (4.55)$$

Injecting this estimate into (4.54) and noting that on  $B(j; G)$  we have to reach at least distance  $3^{k-1}$  yields

$$\begin{aligned} \mathbb{P}_{p_c}(H(3^k; G), |\mathcal{C}_G(0)| \leq A^{-4/3} 9^k) &\leq A^{-4/3} 3^{k+1} \Gamma(3^{k-1}) \mathbb{P}_{p_c}(\partial B(0, 3^{k-1}; G) \neq \emptyset) \\ &\leq A^{-4/3} 3^{k+1} \Gamma(3^{k-1})^2. \end{aligned} \quad (4.56)$$

Combining this with (4.49), (4.52), (4.53) and the induction hypothesis gives us

$$\begin{aligned} \mathbb{P}_{p_c}(H(3^k; G)) &\leq C_\delta A^{2/3} 3^{-k} + A^{-4/3} 3^{k+1} \Gamma(3^{k-1})^2 \\ &\leq C_\delta A^{2/3} 3^{-k} + A^{-4/3} 3^{k+1} \left( \frac{A}{3^{k-1}} \right)^2 \\ &= \frac{3^3 A^{2/3} + C_\delta A^{2/3}}{3^k} \leq \frac{A}{3^k}, \end{aligned} \quad (4.57)$$

thus finishing our induction and the proof by taking the supremum over all subgraphs  $G$ .  $\square$

From Theorem 4.2 we could also follow the continuity of the percolation function, since we see that

$$\theta(p_c) \leq \mathbb{P}_{p_c}(\text{rad}_{\text{int}} \geq n) \leq \frac{C_{\text{in}}}{n}, \quad (4.58)$$

and notice that the right hand side goes to 0 for  $n$  to infinity. However as already seen, this is not the fastest way to get to this statement, since both Theorem 4.2 and Theorem 4.3 rely solely on the triangle condition. This is because we used critical exponents we established in Section 3.1 that hold under the finiteness of the triangle, which already implies the continuity of the percolation function. The statement is also true for long range percolation, given the triangle condition. Currently we know that the theorem holds for  $d \geq 11$  and the proof generalizes this to arbitrary dimensions  $d > 6$  if we have the triangle condition at hand. This is not given at the time but as already mentioned predicted to be true. We can therefore claim with high confidence that  $\rho_{\text{in}} = 1$  for  $d > 6$ .

Also our statement is a bit weaker than the one we had for the “easier” setting of percolation on  $\mathbb{T}_{2d}$  in Proposition 4.1. In contrast, we do not receive a similar asymptotic statement, since the existence of the limit

$$a_{\text{in}} := n \mathbb{P}_{p_c}(\text{rad}_{\text{int}} \geq n) \quad (4.59)$$

remains unknown.

## 4.2 Related Results

For  $d \leq 6$  the results on the intrinsic distance are far worse. The main reason for this is the lack of our critical exponent machinery we used excessively to get nice upper and lower bounds on important objects. Nevertheless we can give the lower bound on the arm event in a substantial more general setting and even completely without the triangle condition. This was shown recently in 2020 by Hutchcroft and the bound holds on more general graphs. It is typical for critical exponents in the bounded-ratio sense for  $\mathbb{Z}^d$  that one bound holds on all locally finite transitive graphs. As already noted, the objects we defined earlier transfer over naturally to more arbitrary graphs.

**Theorem 4.4** (Hutchcroft [28]). *Let  $G = (V, E)$  be an infinite, connected, locally finite, quasi-transitive graph. Then there exist positive constants  $c, C$  and  $\delta$  such that*

$$\mathbb{P}_p(\text{rad}_{\text{int}}(v) \geq n) \geq \frac{c}{n} \exp(-C|p - p_c|n) \quad (4.60)$$

for every  $p \in (p_c - \delta, p_c]$ ,  $n \geq 1$ , and  $v \in V$ .

*Proof.* Fix  $v \in V$ , let  $n \geq 1$ . Then Russo's formula (2.13) yields

$$\frac{d}{dp} \mathbb{P}_p(\text{rad}_{\text{int}}(v) \geq n) = \frac{1}{p} \sum_{e \in E} \mathbb{E}_p \left[ \mathbb{1}(\omega(e) = 1) (\mathbb{1}_{\{\text{rad}_{\text{int}}(v) \geq n\}}(\omega) - \mathbb{1}_{\{\text{rad}_{\text{int}}(v) \geq n\}}(\omega_e)) \right]. \quad (4.61)$$

Assume that  $\mathcal{C}(v)$  has intrinsic radius of at least  $n$  on the configuration  $\omega$  and the edge  $e$  is such that  $\mathcal{C}(v)$  has intrinsic radius of less than  $n$ . This means that the absence of  $e$  blocks all shortest paths of length  $n$  that start in  $v$ . There cannot be more than  $n$  such edges, since  $e$  has to be on a shortest path. Thus with (4.61)

$$\frac{d}{dp} \mathbb{P}_p(\text{rad}_{\text{int}}(v) \geq n) \leq \frac{n}{p} \mathbb{P}_p(\text{rad}_{\text{int}}(v) \geq n). \quad (4.62)$$

Using the good old derivative rule of the logarithm, this inequality can be rewritten as

$$\frac{d}{dp} \log \mathbb{P}_p(\text{rad}_{\text{int}}(v) \geq n) \leq \frac{n}{p}. \quad (4.63)$$

Let  $p \leq q \leq 1$ . We can now integrate this inequality from  $p$  to  $q$  to get

$$\log \mathbb{P}_q(\text{rad}_{\text{int}}(v) \geq n) - \log \mathbb{P}_p(\text{rad}_{\text{int}}(v) \geq n) \leq \frac{n}{p}(q - p), \quad (4.64)$$

what we can rephrase to receive

$$\mathbb{P}_p(\text{rad}_{\text{int}}(v) \geq n) \geq \mathbb{P}_q(\text{rad}_{\text{int}}(v) \geq n) \exp \left( -\frac{q - p}{p} n \right). \quad (4.65)$$

Let  $q = 1$  and for  $0 < \delta < p_c$  choose  $C$  such that  $1/p < C$  holds for  $p \in (p_c - \delta, p_c]$ . For these values of  $p$  we can then estimate

$$\mathbb{P}_p(\text{rad}_{\text{int}}(v) \geq n) \geq \exp \left( -\frac{1 - p}{p} n \right) \geq \frac{1}{n} \exp(-C|p - 1|n) \quad (4.66)$$

for every  $n \geq 1$  and the claim follows for the uninteresting case of  $p_c = 1$ . For  $p_c < 1$  we use the analogue to  $\beta = 1$  for this setting: There exists a positive constant  $c > 0$  such that  $\mathbb{P}_p(|\mathcal{C}(v)| = \infty) \geq c(p - p_c)$  for all  $p \geq p_c$ . For a proof of this claim, see [13, Theorem 1.1]. Thus using (4.65), and since  $q \geq p_c$  we receive

$$\mathbb{P}_p(\text{rad}_{\text{int}}(v) \geq n) \geq c(q - p_c) \exp \left( -\frac{q - p}{p} n \right) \quad (4.67)$$

for every  $0 \leq p \leq p_c \leq q \leq 1$ . Letting  $q - p_c = n^{-1}$  finishes the proof.  $\square$

The theorem gives us a lower bound not only in the critical regime, but also for *slightly subcritical* values, meaning that  $0 < p_c - p \ll 1$ . Since  $\mathbb{Z}^d$  trivially fulfills all the conditions of Theorem 4.4 for every  $d$ , we can use it with  $p = p_c, v = 0$  and get the existence of a constant  $c_{\text{in}} > 0$  such that

$$\mathbb{P}_{p_c}(\text{rad}_{\text{int}} \geq n) \geq \frac{c_{\text{in}}}{n}. \quad (4.68)$$

This means that with our coupling and second moment proof we now saw three alternatives to show the lower bound for  $\rho_{\text{in}} = 1$ :

1. The second moment proof is limited to the euclidean case, since there we used the intrinsic growth property (4.26), whose proof used the critical exponent  $\eta$ , and thus an euclidean norm.
2. The coupling approach also goes through for more general graphs, since we only needed  $\beta = 1$ . However for this to hold, we utilized the triangle condition.
3. The proof of Theorem 4.4 is even more general, since it does not make use of the triangle condition.

With this more general statement we can give a bound on  $\rho_{\text{in}}$  for  $d \geq 1$ , since this implies that  $\rho_{\text{in}} \geq 1$ , if  $\rho_{\text{in}}$  exists. We are not aware of any results studying a corresponding upper bound in dimensions  $2 \leq d \leq 6$ , but we would be surprised if  $\rho_{\text{in}} = 1$  holds in these cases, since critical exponents are normally not constant over all dimensions.

If we further assume a boundedness condition on  $p_c$ , it can be shown that (4.60) even holds for *slightly supercritical* values. For that we define for a countable graph  $G = (V, E)$  the *two point matrix*  $T_p$  as  $T_p(u, v) = \mathbb{P}_p(u \longleftrightarrow v)$ . With this we can set

$$p_{2 \rightarrow 2} = p_{2 \rightarrow 2}(G) := \sup\{p \in [0, 1] : \|T_p\|_{2 \rightarrow 2} < \infty\}, \quad (4.69)$$

where  $\|\cdot\|_{2 \rightarrow 2}$  is the normal operator norm for  $L^2(V) \rightarrow L^2(V)$ , given by

$$\|T_p\|_{2 \rightarrow 2} = \sup\{\|T_p f\|_2 : f \in L^2(V), \|f\|_2 = 1\}. \quad (4.70)$$

We now say that  $G$  satisfies the  $L^2$  *boundedness condition* if  $p_c < p_{2 \rightarrow 2}$ . When a graph has this property [28, Proposition 4.5] establishes that

$$\mathbb{P}_p(\text{rad}_{\text{int}}(v) \geq n) \geq \frac{c}{n} \exp(-C|p - p_c|n) \quad (4.71)$$

also for  $0 < p - p_c \ll 1$ . However we stress the point that  $p_c = p_{2 \rightarrow 2}$  on  $\mathbb{Z}^d$  for all dimensions.

For completeness we give an upper bound for general graphs also holding in the subcritical regime. However we still need a finite triangle here. Together with Theorem 4.4 this shows  $\rho_{\text{in}} = 1$  under the triangle condition.

**Theorem 4.5** (Hutchcroft [28]). *Let  $G = (V, E)$  be an infinite, connected, locally finite, quasi-transitive graph and suppose that  $\Delta_{p_c} < \infty$ . Then there exist positive constants  $c$  and  $C$  such that*

$$\mathbb{P}_{p_c}(\text{rad}_{\text{int}}(v) \geq n) \leq \frac{C}{n} \exp(-c|p - p_c|n) \quad (4.72)$$

for every  $n \geq 1$ ,  $0 \leq p \leq p_c$ , and  $v \in V$ .

A lot of times in the literature the intrinsic distance of the shortest path in a box-crossing is studied, also called *crossing event*. For that, let  $H_n$  be the event that we have a horizontal open path inside  $[-n, n]^d$ . This means we want to connect the left side  $-n \times [-n, n]^{d-1}$  with the right side  $n \times [-n, n]^{d-1}$ . Let  $S_n$  be the minimal intrinsic distance of all such paths. From [17] it follows that for  $d \geq 2$

$$\frac{S_n}{n} \rightarrow \text{constant a.s.} \quad (4.73)$$

Specifically in dimensions  $3 \leq d \leq 6$  not much progress seems to have been made for understanding the intrinsic distance. Yet there exist some partial or similar results in the two dimensional case [9]. On  $H_n$  we can additionally define  $L_n$  to be the intrinsic length of the lowest open crossing, if we restrict ourself to dimension two. In [31] it was asked, whether for  $d = 2$  and all  $\epsilon > 0$

$$\lim_{n \rightarrow \infty} \mathbb{P}_{p_c} \left( \frac{S_n}{L_n} > \epsilon \mid H_n \right) \rightarrow 0. \quad (4.74)$$

This is just convergence to 0 in probability with respect to the conditional measure. The question was answered by Damron, Hanson and Sosoe [10], by proving the convergence.

In the physics literature it is believed that there exists a critical exponent  $d_{\min}$  such that

$$\mathbb{E}_{p_c}[S_n \mid H_n] \sim n^{d_{\min}}, \quad (4.75)$$

where the exact meaning of  $\sim$  still needs to be determined and there is no generally accepted prediction for an exact value. In the survey of Schramm [44, Problem 3.3] this is stated as an important open problem. It is however not yet clear how to connect the intrinsic distance to conformal invariant quantities. Thus the usage of SLE<sup>3</sup> analysis, which would be the standard approach, seems not applicable here. Very recently a first non-trivial upper bound on  $d_{\min}$  where  $\sim$  was meant in the bounded-ratio sense was shown in [11]. They were able to prove that  $d_{\min} \leq 4/3$  holds. There it is also stated that for two dimensions there exists  $\delta > 0$  and a constant  $C > 0$  such that for each  $n \geq 1$

$$\mathbb{E}_{p_c}[S_n \mid H_n] \leq C n^{-\delta} \mathbb{E}_{p_c}[L_n \mid H_n]. \quad (4.76)$$

The authors of [11] summarize today's research progress in two dimensions with: "The existence and determination of an exponent for the chemical distance in any two-dimensional short-range critical percolation model is thus far out of reach of current methods. [...] Apart from its mathematical appeal, further progress on the chemical distance is a significant obstacle to analyzing random walks on low-dimensional critical percolation clusters. [...] In the low-dimensional, critical case, the chemical distance is not well understood, even at the physics level of rigor."

We finish this part with an estimate holding for critical percolation in high dimensions. Similarly to  $S_n$  we could define  $\tilde{S}_n$  to be the intrinsic distance of the origin to  $[-n, n]^d$ . So we seek connections from 0 to the boundary of

$$\Lambda_n := \{x : \|x\|_{\infty} \leq n\}, \quad (4.77)$$

---

<sup>3</sup>SLE abbreviates stochastic Loewner evolution, a one parameter family of random curves. They appear as the limit of paths in statistical physics models as the mesh of the grid where the model is defined goes to zero. These quantities are used excessively to derive results in two dimensions. For further descriptions and more references we recommend [44].

and measure the shortest path. If we condition on the event  $\{0 \longleftrightarrow \partial\Lambda_n\}$ , we can use (4.32) to follow that  $\tilde{S}_n$  should be of order  $n^2$ . In a recent work Chatterjee, Hanson and Sosoe [8] proved that the normalized quantity  $n^{-2}\tilde{S}_n$  has a lower tail that decays exponentially.

**Theorem 4.6** (Exponential decay of  $\tilde{S}_n$  [8]). *Under the triangle condition there is a constant  $c > 0$  such that for any  $\lambda > 0$*

$$\mathbb{P}_{p_c}(\tilde{S}_n \leq \lambda n \mid 0 \longleftrightarrow \partial\Lambda_n) \leq \exp(-c\lambda^{-1}), \quad (4.78)$$

and there is a constant  $C > 0$  such that for all  $\lambda \geq Cn^{-1}$

$$\mathbb{P}_{p_c}(\tilde{S}_n \leq \lambda n \mid 0 \longleftrightarrow \partial\Lambda_n) \geq \exp(-C\lambda^{-1}). \quad (4.79)$$

### 4.3 Excursion: The Extrinsic Arm

Thus far we considered the probability of a connection of the origin to the boundary of a ball. This ball was defined with respect to the intrinsic distance but we could also study this with a different metric. One possibility is to use a metric on  $\mathbb{Z}^d$ , like the one induced by the  $\ell^\infty$ -norm, which is by far the most commonly used here. This would then be an extrinsic distance on  $\mathbb{Z}^d$ . In the euclidean setting there actually exists more research output concerning this compared to the intrinsic version.

With this metric we want to examine the probability of a connection of the origin to the boundary of  $\Lambda_n$ , which is just the ball with respect to the  $\|\cdot\|_\infty$  norm. As before, define the *extrinsic one-arm critical exponent*  $\rho_{\text{ex}} > 0$  by

$$\mathbb{P}_{p_c}(0 \longleftrightarrow \partial\Lambda_n) \asymp n^{-1/\rho_{\text{ex}}}, \quad n \rightarrow \infty. \quad (4.80)$$

The most prominent case where a connection to  $\partial\Lambda_n$  was used, is found in the proof of the uniqueness of the phase transition, see Theorem 2.2. For that the following decay statement can be utilized.

**Theorem 4.7** (Exponential decay in the subcritical phase [14]). *For  $d \geq 2$  and every  $p < p_c$ , there exists  $c = c(d, p) > 0$  such that for all  $n \geq 1$*

$$\mathbb{P}_p(0 \longleftrightarrow \partial\Lambda_n) \leq \exp(-cn). \quad (4.81)$$

We can now follow Grimmett [18, Section 5.2] and define  $M := \max\{n : 0 \longleftrightarrow \partial\Lambda_n \text{ occurs}\}$  and realize that if  $p < p_c$  we get  $\mathbb{P}_p(M < \infty) = 1$ . Since for  $n \geq 1$  under  $\{M = n\}$  the cluster size cannot be larger than  $|\Lambda_n|$ , we can use the exponential decay and estimate for  $p < p_c$

$$\begin{aligned} \mathbb{E}_p|\mathcal{C}(0)| &= \sum_n \mathbb{E}_p[|\mathcal{C}(0)| \mid M = n] \mathbb{P}_p(M = n) \\ &\leq \sum_n |\Lambda_n| \mathbb{P}_p(0 \longleftrightarrow \partial\Lambda_n) \\ &\leq \sum_n (n+1)^d e^{-cn} < \infty. \end{aligned} \quad (4.82)$$

As an analogue to Theorem 4.3 where we gave bounds on the intrinsic volume, we could be interested in the extrinsic counterpart. The following also holds for a variety of long-range spread-out models.

**Theorem 4.8** (Extrinsic expected volume [23]). *For  $d$  high enough*

$$\mathbb{E}_{p_c} |\Lambda_n \cap \mathcal{C}(0)| \asymp n^2. \quad (4.83)$$

It seems like 2009 was a big year for arm exponents. Kozma and Nachmias published results for both the intrinsic and extrinsic case during that period:

**Theorem 4.9** ( $\rho_{\text{ex}} = 1/2$  [34]). *Consider percolation under the triangle condition. Then there exist constants  $0 < c_{\text{ex}} < C_{\text{ex}} < \infty$ , such that for all  $n \in \mathbb{N}$*

$$\frac{c_{\text{ex}}}{n^2} \leq \mathbb{P}_{p_c}(0 \longleftrightarrow \partial\Lambda_n) \leq \frac{C_{\text{ex}}}{n^2}. \quad (4.84)$$

*So the extrinsic critical arm exponent exists in the bounded-ratio sense and takes on his mean field value  $\rho_{\text{ex}} = 1/2$ .*

In [8] the subcritical case was studied. It was shown that under  $\eta = 0$  there exist a dimension dependent constant  $C > 0$ , such that for all  $n \in \mathbb{N}$  and  $p \in (0, p_c]$

$$\frac{1}{C} n^{-2} \exp(-Cn\sqrt{p_c - p}) \leq \mathbb{P}_p(0 \longleftrightarrow \partial\Lambda_n) \leq Cn^{-2} \exp\left(-\frac{n\sqrt{p_c - p}}{C}\right). \quad (4.85)$$

Since we need at least  $n$  steps for a path to the boundary of the cube with side length  $n$ , we have  $\{0 \longleftrightarrow \Lambda_n\} \subset \{0 \longleftrightarrow B(n)\}$  and it is not surprising that the leading order compared to the intrinsic case is therefore different. Likewise we have  $|B(n)| \leq |\Lambda_n \cap \mathcal{C}(0)|$  and for most configurations this inequality is strict, what is then resulting in the mismatched factors of  $n$ , resp.  $n^2$ . Again, the proof of the lower bound in (4.84) can be done by the second moment method (4.40), this time applied to the random variable

$$X = |\{x \in \Lambda_{2n} \setminus \Lambda_n : 0 \longleftrightarrow x\}|. \quad (4.86)$$

The proof of the upper bound is significantly longer, therefore we will skip it here.

Interestingly we can use the extrinsic critical exponents  $\rho_{\text{ex}}$  and  $\eta$  to show that  $d_c \geq 6$ , which is a priori not clear if we just know the critical exponents for  $d > 6$  but have no exact values for dimension  $6 \leq d$ .

**Corollary 4.10** ( $d_c \geq 6$  [25]). *The mean-field values  $\rho_{\text{ex}} = 1/2$  and  $\eta = 0$  imply that the upper critical dimension  $d_c$  satisfies  $d_c \geq 6$ .*

*Proof.* We will show that if  $\rho_{\text{ex}} = 1/2$  and  $\eta = 0$  holds,  $d \geq 6$  follows. This implies  $d_c \geq 6$ . Let  $n \in \mathbb{N}$  and  $e_1 = (1, 0, \dots, 0) \in \mathbb{Z}^d$ . Note that if  $0 \longleftrightarrow 2ne_1$ ,  $0$  is connected to  $\partial\Lambda_n$  and  $2ne$  is connected to the shifted boundary of the box  $2ne + \partial\Lambda_n$ . Figure 4.2 makes this visual. With this observation we can estimate

$$\tau_{p_c}(2ne_1) = \mathbb{P}_{p_c}(0 \longleftrightarrow 2ne_1) \leq \mathbb{P}_{p_c}(\{0 \longleftrightarrow \partial\Lambda_n\} \cap \{2ne_1 \longleftrightarrow (2ne_1 + \partial\Lambda_n)\}). \quad (4.87)$$

Note that the two events  $\{0 \longleftrightarrow \partial\Lambda_n\}$  and  $\{2ne_1 \longleftrightarrow (2ne_1 + \partial\Lambda_n)\}$  both occur inside disjoint boxes, so they are also disjoint in a probabilistic matter. Since for both events we need a connection from the center to the boundary of a box, we can use the translation invariance of the measure to conclude that both events have the same probability. This yields

$$\tau_{p_c}(2ne_1) \leq \mathbb{P}_{p_c}(0 \longleftrightarrow \partial\Lambda_n)^2. \quad (4.88)$$

We now use that  $\eta = 0$  to get the existence of a constant  $c > 0$  such that  $cn^{2-d} \leq \tau_{pc}(2ne_1)$ . On the other hand we utilize  $\rho_{\text{ex}} = 1/2$  to get  $\mathbb{P}_{pc}(0 \longleftrightarrow \partial\Lambda_n)^2 \leq C_{\text{ex}}^2/n^4$ . Combining these estimates with (4.88) results in

$$cn^{2-d} \leq \frac{C_{\text{ex}}^2}{n^4}, \quad (4.89)$$

and this can only hold for all  $n \geq 1$  if  $d \geq 6$ .  $\square$

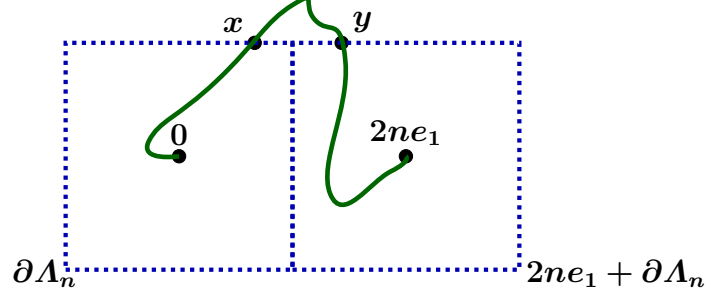


Figure 4.2: If we have a path from 0 to  $2ne_1$  we can connect both boxes to the corresponding center, by connecting 0 with  $x$  and  $2ne_1$  with  $y$ .

One might now want to calculate the corresponding exponent  $\rho_{\text{ex}}$  for the tree and compare them. This is however not feasible, since the intrinsic and extrinsic distances we defined are exactly the same on a tree. If we want to get to the boundary of a ball with radius  $n$  on a tree, we want to have a connection to a vertex which has distance  $n$ . Since we have no loops these vertices always have graph distance  $n$ . Moreover speaking of a euclidean norm for these graphs is troublesome, since we already pointed out that there is no euclidean embedding for the regular tree. Thus we do not get a new exponent for a tree.

In contrast to the intrinsic setting, there exist precise results for the exponent in dimension 2. In [36] it is shown that  $\rho_{\text{ex}} = 48/5$  for site percolation on the planar triangular lattice, relying on the SLE results of Smirnov [46]. Like before, these results are predicted to hold on any planar lattice. Again we are not aware of any results focusing on dimensions  $3 \leq d \leq 6$ .

In the same paper where they showed  $\rho_{\text{ex}} = 1/2$  in high dimensions, Nachmias and Kozma also considered not one path to the sphere, but the occurrence of multiple disjoint paths. They showed that we can scale  $\rho_{\text{ex}}$  by  $l$  to receive an arm exponent for  $l$  paths.

**Theorem 4.11** (Multiple extrinsic arms [34]). *Under the triangle condition and every integer  $l \geq 1$  there exists a constant  $K = K(l, d)$  such that for any  $n \in \mathbb{N}$  and for all vertices  $v_1, \dots, v_l \in \Lambda_{n/2}$  satisfying  $|v_i - v_j| \geq K$  for any  $i \neq j$  we have*

$$\mathbb{P}_{pc}(\{v_1 \longleftrightarrow \Lambda_n\} \circ \dots \circ \{v_l \longleftrightarrow \Lambda_n\}) \asymp n^{-2l}. \quad (4.90)$$

Note that the upper bound of (4.90) is completely trivial, if we use the BK inequality together with  $\rho_{\text{ex}} = 1/2$ . To conclude the proof we need an estimate of the same order in the other direction, so some kind of inverse BK statement. As it happens, this derivation is heavily inspired by the proof of the classical BK inequality for increasing events. See Section 5.2.1 for a presentation of the key ideas.

The natural question we now have is if we also get a theorem like this for the intrinsic radii. This shall be the motivation for our next chapter.

# Chapter 5

## Multiple Arms

We saw in detail how the one-arm probability behaves. A natural extension now would be to consider not one arm, but the occurrence of multiple arms at the same time, like the ones we identified for the extrinsic distance. For that define for  $v_1, \dots, v_l \in \mathbb{Z}^d$  the *intrinsic  $l$ -arm critical exponent*  $\rho_l > 0$

$$\mathbb{P}_{p_c}(\{\text{rad}_{\text{int}}(v_1) \geq n\} \circ \dots \circ \{\text{rad}_{\text{int}}(v_l) \geq n\}) \asymp n^{-1/\rho_l}. \quad (5.1)$$

We saw that the upper bound of the BK inequality is in a certain sense sharp for the extrinsic multi-arm exponent. Is this also the case for numerous intrinsic arms? We are not aware of any published results answering this question for  $\mathbb{Z}^d$ . Note that the sites  $v_1, \dots, v_l$  on the event  $\{\text{rad}_{\text{int}}(v_1) \geq n\} \circ \dots \circ \{\text{rad}_{\text{int}}(v_l) \geq n\}$  can be in the same cluster, but the corresponding shortest paths cannot cross. Furthermore we have to include edges in our witness set that would create shortcuts and therefore destroy long arms.

Very recently Hutchcroft showed that  $\rho_l = 1/l$  is the correct value under the assumption that  $p_c < p_{2 \rightarrow 2}$ , what we remember as the  $L^2$  boundedness condition from around (4.70).

**Theorem 5.1** ( $\rho_l = 1/l$  under the  $L^2$  boundedness condition [29]). *Let  $G = (V, E)$  be a connected, locally finite, transitive graph with  $p_c < p_{2 \rightarrow 2}$ . For each  $l \geq 2$  there exists a finite constant  $K = K(l) > 0$  such that*

$$\mathbb{P}_{p_c}(\{\text{rad}_{\text{int}}(v_1) \geq n\} \circ \dots \circ \{\text{rad}_{\text{int}}(v_l) \geq n\}) \asymp n^{-1/\rho_l} \quad (5.2)$$

*for every  $v_1, \dots, v_l \in V$  if  $d(v_i, v_j) \geq K$  for every  $1 \leq i < j \leq l$ .*

Above  $d(\cdot, \cdot)$  denotes the deterministic graph distance on the non-percolated fully connected graph.

So is this already the statement we are after? It is conjectured that  $p_c < p_{2 \rightarrow 2}$  holds on every connected, locally finite, nonamenable quasi-transitive graph. A graph  $G = (V, E)$  is said to be nonamenable if there exists a constant  $c > 0$  such that for every finite subset  $W \subset V$  we have  $|\partial W|/|W| > c$ . Otherwise the graph is *amenable*.

We already saw an example for a nonamenable graph in Figure 3.1. To see that the  $d$ -regular tree is nonamenable for  $d \geq 3$ , we first bound the average degree of a finite subgraph  $(W, E(W))$  of  $G$  with  $n$  sites: If we use that a tree with  $n$  vertices has  $n - 1$  edges, we get

$$\frac{1}{n} \sum_{w \in W} \deg(w) = \frac{1}{n} 2|E(W)| = \frac{1}{n} 2(n - 1) = 2 - \frac{2}{n} \leq 2. \quad (5.3)$$

With that we compute

$$|\partial W| = d|W| - 2|E(W)| = \left(d - \frac{2|E(W)|}{|W|}\right)|W| \geq (d-2)|W|, \quad (5.4)$$

implying nonamenability. Unfortunately, Hutchcroft's result does not transfer over to our setting, since  $\mathbb{Z}^d$  is amenable and  $p_c = p_{2 \rightarrow 2}$  for amenable graphs. To see that  $\mathbb{Z}^d$  is amenable, take  $W = [0, n]^d \cap \mathbb{Z}^d$  and compute that  $|W| = n^d$  but  $|\partial W| = n^d - (n-1)^d \approx n^{d-1}$ , resulting in

$$\frac{|\partial W|}{|W|} = \frac{n^d - (n-1)^d}{n^d} \approx \frac{n^{d-1}}{n^d} \xrightarrow{n \rightarrow \infty} 0. \quad (5.5)$$

In his proof the fact was used that on transitive nonamenable graphs in the supercritical regime we have infinitely many disjoint infinite clusters. As we know this is not the case for  $\mathbb{Z}^d$ . Nonetheless we will try to use two of his lemmas to get a similar statement. To do so we first have to introduce and study a new random object.

## 5.1 Magnetization and Ghost Fields

We will now define a special kind of random subset of our vertex set. It was first introduced by Aizenman and Barsky [1] and was used to prove the uniqueness of the phase transition. Later on we will see how to utilize it.

For an intensity  $h \geq 0$  we define a *ghost field*  $\mathcal{G}$  to be a random subset of  $\mathbb{Z}^d$ , independent of our percolation configuration. Include every vertex  $x \in \mathbb{Z}^d$  independently with probability  $1 - e^{-h}$ . Vertices included in the ghost field are sometimes called *green*. We can think of this as a combination of site and bond percolation and write  $\mathbb{P}_{p,h}$  for the joint law. Figure 5.1 shows how such a random object might look like. Define the *magnetization*  $M_{p,h}$  as the probability, that the cluster of the origin contains at least one green vertex. Equivalently we could say that 0 (alternatively by translation invariance any other reference point could be chosen) is connected to the ghost field. That is

$$M_{p,h} = \mathbb{P}_{p,h}(0 \longleftrightarrow \mathcal{G}) = \mathbb{P}_{p,h}(\mathcal{C}(0) \cap \mathcal{G} \neq \emptyset). \quad (5.6)$$

Since  $|\mathcal{C}(0) \cap \mathcal{G}|$  conditioned on  $\mathcal{C}(0)$  is  $\text{Bin}(\mathcal{C}(0), 1 - e^{-h})$  distributed, we get the useful expression

$$M_{p,h} = \mathbb{P}_{p,h}(|\mathcal{C}(0) \cap \mathcal{G}| \geq 1) = 1 - \mathbb{P}_{p,h}(|\mathcal{C}(0) \cap \mathcal{G}| = 0) = \mathbb{E}_p \left[ 1 - e^{-h|\mathcal{C}(0)|} \right]. \quad (5.7)$$

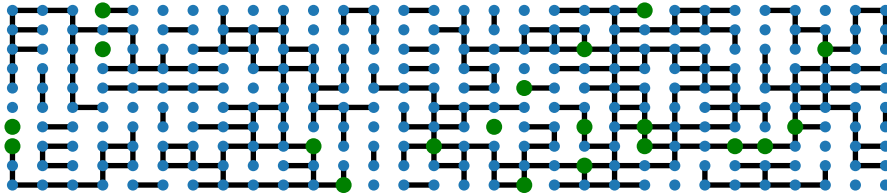


Figure 5.1: How a combination of site and bond percolation could look like. Vertices included in the ghost field are marked green. The graphic shows a subset with size  $10 \times 60$ , bond probability  $p = 0.5$  and intensity  $h = 0.08$ .

We will use this equality numerous times. Another interesting fact about the magnetization that is worth knowing is

$$\lim_{h \searrow 0} M_{p,h} = \theta(p). \quad (5.8)$$

The following lemma gives a very powerful explicit lower bound on the magnetization. To do so we define a truncated version of the susceptibility by

$$\chi^f(p) = \mathbb{E}_p \left[ |\mathcal{C}(0)| \mathbb{1}_{\{|\mathcal{C}(0)| < \infty\}} \right]. \quad (5.9)$$

**Lemma 5.2** (Magnetization lower bound [25]). *Let  $p \in (0, 1)$  be such that  $\chi^f(p) = \infty$ , and  $h \in (0, 1)$ . Then we have*

$$M_{p,h} \geq \sqrt{\frac{h}{K}}, \quad (5.10)$$

with  $K = 1 + \frac{2dp}{1-p}$ .

Note that we can use this lemma with  $p = p_c$ , since in high dimensions the triangle diagram is finite. Therefore we can use that  $\theta(p_c) = 0$ , implying  $\chi^f(p_c) = \chi(p_c) = \infty$ . Let now  $G_1, \dots, G_l$  be ghost fields with intensities  $(h_1, \dots, h_l) = \mathbf{h} \in (0, \infty)^l$  on some percolation configuration  $\omega$  and write  $\mathbb{P}_{p,\mathbf{h}}$  for their joint law. We will introduce two lemmas that are combined later. The first lemma gives us a lower bound on the probability that vertices in disjoint clusters are connected to independent ghost fields with possibly different intensities.

**Lemma 5.3** (Hutchcroft [29]). *Let  $l \geq 2$  and  $v_1, \dots, v_l$  be vertices of  $\mathbb{Z}^d$ . Then*

$$\begin{aligned} & \mathbb{P}_{p,\mathbf{h}}(\mathcal{C}(v_1), \dots, \mathcal{C}(v_l) \text{ are disjoint and } v_i \longleftrightarrow \mathcal{G}_i \text{ for every } 1 \leq i \leq n) \\ & \geq \mathbb{P}_{p,\mathbf{h}}(\{v_1 \longleftrightarrow \mathcal{G}_1\} \circ \dots \circ \{v_l \longleftrightarrow \mathcal{G}_l\}) - l(l-1) \left( \prod_{i=1}^l M_{p,h_i} \right) \sup_{1 \leq i < j \leq l} \tau_p^{*3}(v_i, v_j) \end{aligned} \quad (5.11)$$

for every  $0 < p < 1$  and  $\mathbf{h} \in (0, \infty)^l$ .

*Proof.* Let

$$\mathcal{A} = \{\mathcal{C}(v_1), \dots, \mathcal{C}(v_l) \text{ are disjoint and } v_i \longleftrightarrow \mathcal{G}_i \text{ for every } 1 \leq i \leq n\} \quad (5.12)$$

and

$$\mathcal{B} = \{v_1 \longleftrightarrow \mathcal{G}_1\} \circ \dots \circ \{v_l \longleftrightarrow \mathcal{G}_l\}. \quad (5.13)$$

It is obvious that  $\mathcal{A} \subseteq \mathcal{B}$ , so we can use elementary properties of probability theory and rewrite

$$\mathbb{P}_{p,\mathbf{h}}(\mathcal{B}) = \mathbb{P}_{p,\mathbf{h}}(\mathcal{B} \cap \mathcal{A}) + \mathbb{P}_{p,\mathbf{h}}(\mathcal{B} \cap \mathcal{A}^c) = \mathbb{P}_{p,\mathbf{h}}(\mathcal{A}) + \mathbb{P}_{p,\mathbf{h}}(\mathcal{B} \setminus \mathcal{A}). \quad (5.14)$$

Thus now assume that  $\mathcal{B} \setminus \mathcal{A}$  occurs. Since this means that  $\mathcal{B}$  holds, there exist edge disjoint open paths  $\gamma_1, \dots, \gamma_l$  connecting every  $v_i$  to their respective ghost field  $\mathcal{G}_i$  for  $1 \leq i \leq l$ . On the other hand  $\mathcal{A}$  does not occur, so the clusters are not disjoint. This means there exists at least one path  $\gamma$  that connects two of our sites  $v_1, \dots, v_l$ . Assume the connection is in between  $v_{j_1}$  and  $v_{j_2}$  for  $1 \leq j_1 < j_2 \leq l$ . Then let  $\gamma'$  be the part of  $\gamma$  that does not use any vertex of  $\gamma_{j_1}$  and  $\gamma_{j_2}$ . With this we can identify  $w_1$  and  $w_2$  as the last common site of  $\gamma'$  and  $\gamma_{j_1}$ , respectively  $\gamma_{j_2}$ . Figure 5.2 makes it visual. With the choices

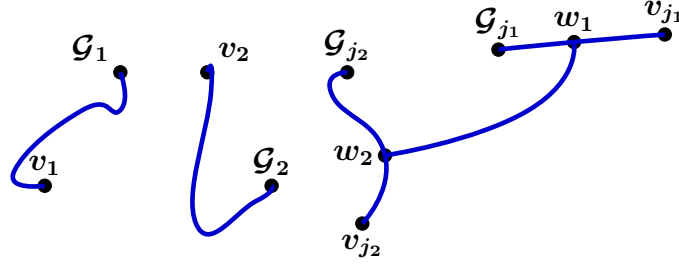


Figure 5.2: How a configuration of  $\mathcal{B} \setminus \mathcal{A}$  for  $l = 4$  might look like. With  $\mathcal{G}_1$  we refer to an arbitrary green vertex of  $\mathcal{G}_1$ .

of  $j_1, j_2, w_1$  and  $w_2$  the events  $\{v_{j_1} \longleftrightarrow w_1\}$ ,  $\{v_{j_2} \longleftrightarrow w_2\}$ ,  $\{w_1 \longleftrightarrow w_2\}$ ,  $\{w_1 \longleftrightarrow \mathcal{G}_{j_1}\}$ , and  $\{w_2 \longleftrightarrow \mathcal{G}_{j_2}\}$  all occur disjointly. Additionally the events  $\{v_i \longleftrightarrow \mathcal{G}_i\}$  for  $i \notin \{j_1, j_2\}$  also occur disjointly from these events and from each other. Using this observations we apply the BK inequality (2.8) and sum over all possible values of  $j_1, j_2, w_1$ , and  $w_2$ , to obtain

$$\begin{aligned} \mathbb{P}_{p, \mathbf{h}}(\mathcal{B} \setminus \mathcal{A}) &\leq \sum_{w_1, w_2 \in \mathbb{Z}^d} \sum_{j_1=1}^l \sum_{j_2 \neq j_1} \tau_p(v_{j_1}, w_1) \tau_p(w_1, w_2) \tau_p(w_2, v_{j_2}) \prod_{i=1}^l M_{p, h_i} \\ &\leq l(l-1) \sup_{1 \leq i < j \leq l} \tau_p^{*3}(v_i, v_j) \left( \prod_{i=1}^l M_{p, h_i} \right). \end{aligned} \quad (5.15)$$

Together with (5.14) this finishes the proof.  $\square$

The disjoint occurrence probability on the right hand side of (5.11) can be estimated from below by the following inverse BK statement. Since the inequality is in the opposite direction of the BK inequality, the term on the right hand side of (5.16) can be interpreted as a penalty. The bound becomes meaningful once the *bubble diagram*  $\tau_p^{*2}$  is sufficiently small.

**Lemma 5.4** (Hutchcroft [29]). *Let  $l \geq 2$  and  $v_1, \dots, v_l$  be vertices of  $\mathbb{Z}^d$ . Then*

$$\begin{aligned} &\mathbb{P}_{p, \mathbf{h}}(\{v_1 \longleftrightarrow \mathcal{G}_1\} \circ \dots \circ \{v_l \longleftrightarrow \mathcal{G}_l\}) \\ &\geq \prod_{i=1}^l M_{p, h_i}(v) - \frac{8d}{p^2} \binom{l-1}{2} \left( \prod_{i=1}^l M_{p, h_i} \right) \sup_{1 \leq i < j \leq l} \tau_p^{*2}(v_i, v_j) \end{aligned} \quad (5.16)$$

for every  $0 < p < 1$  and  $\mathbf{h} \in (0, \infty)^l$ .

We postpone an outline of the proof to Section 5.2.1.

## 5.2 The Multi-Arm Exponent

Assuming existence we first we give *mean-field bounds* on  $\rho_l$ , i.e. an inequality on a critical exponent in the mean-field regime.

**Lemma 5.5** (Bounds on  $\rho_l$ ). *If  $\rho_l$  exists and  $\Delta_{p_c} < \infty$  holds we have*

$$\frac{1}{2l} \leq \rho_l \leq \frac{1}{l}. \quad (5.17)$$

*Proof.* Using  $\rho_{\text{in}} = 1$  and the BK-Reimer inequality (2.9) we can easily compute

$$\mathbb{P}_{p_c}(\{\text{rad}_{\text{int}}(v_1) \geq n\} \circ \cdots \circ \{\text{rad}_{\text{int}}(v_l) \geq n\}) \leq \left( \mathbb{P}_{p_c}(\text{rad}_{\text{int}} \geq n) \right)^l \leq \frac{C_{\text{in}}^l}{n^l}. \quad (5.18)$$

So  $\rho_l$  should be at most  $1/l$ , assuming  $\rho_l$  to be existent. If we use the results on extrinsic multiple arms, computing a lower bound of a different order is also easy. For that let  $v_1, \dots, v_l \in \Lambda_n$  be such that  $|v_i - v_j| > K$  for each  $1 \leq i < j \leq l$ , where  $K$  is from Theorem 4.11. This means that the conditions of Theorem 4.11 are fulfilled and a connection of one of the sites to  $\partial\Lambda_{2n}$  will need at least  $n$  steps. This gives us the existence of a constant  $c = c(l, d) > 0$  such that under the triangle condition

$$\begin{aligned} & \mathbb{P}_{p_c}(\{\text{rad}_{\text{int}}(v_1) \geq n\} \circ \cdots \circ \{\text{rad}_{\text{int}}(v_l) \geq n\}) \\ & \geq \mathbb{P}_{p_c}(\{v_1 \longleftrightarrow \partial\Lambda_{2n}\} \circ \cdots \circ \{v_l \longleftrightarrow \partial\Lambda_{2n}\}) \\ & \geq cn^{-2l}, \end{aligned} \quad (5.19)$$

implying another mean-field bound  $\rho_l \geq 1/(2l)$ . □

After all the necessary preliminary work we have all the tools we need and are finally able to state our main claim. Since our mean-field model is nonamenable and from Theorem 5.1 we received  $\rho_l = 1/l$  on  $\mathbb{T}_{2d}$ , we expect that to be also true for high dimensions on  $\mathbb{Z}^d$ . The upper bound for this was already computed in (5.18).

**Conjecture** ( $\rho_l = 1/l$ ). *If the triangle condition holds, for every integer  $l \geq 2$  there exist constants  $0 < c_l < C_l < \infty$  such that*

$$\frac{c_l}{n^l} \leq \mathbb{P}_{p_c}(\{\text{rad}_{\text{int}}(v_1) \geq n\} \circ \cdots \circ \{\text{rad}_{\text{int}}(v_l) \geq n\}) \leq \frac{C_l}{n^l}, \quad (5.20)$$

*for every  $v_1, \dots, v_l$  that have at least distance  $K(l, d) > 0$  from each other. So the  $l$ -arm critical exponent  $\rho_l$  exist in the bounded-ratio sense and takes on the mean-field value  $\rho_l = 1/l$ .*

Always keep in mind that  $c_l$  and  $C_l$  can depend on the dimension  $d$  and the number of paths  $l$ . Also note that as in Theorem 4.11 we have to choose the vertices with some distance  $K(l, d) > 0$  from each other. Surely if we would not do this it would be absolutely impossible to have a certain number of disjoint paths with at least length  $n$ . For instance, we can just have at most  $2d$  disjoint connections emerging from a fixed vertex, so setting all  $v_1, \dots, v_l$  to the origin or really close together would not work out. It is important however that  $K(l, d)$  should not depend on  $n$ . Indeed we could just put the sites at extrinsic distance  $2n$  from each other and it would be impossible for the intrinsic arms to use the same edges. Therefore the statement would lose its significance. The whole trouble now

is the derivation of a lower bound of same order, since we already saw in (5.18) that the upper bound follows directly from the BKR inequality and the one arm critical exponent. We can only give a proof idea but later we will recognize that the sign of the lower bound is problematic.

**Proposition 5.6.** *Under the triangle condition*

$$\frac{1}{n^l} \left( \frac{1}{2K^{l/2}} - G \right) \leq \mathbb{P}_{p_c}(\{\text{rad}_{\text{int}}(v_1) \geq n\} \circ \cdots \circ \{\text{rad}_{\text{int}}(v_l) \geq n\}), \quad (5.21)$$

where  $K$  is from Theorem 5.2 and

$$G := (C_{\text{in}} + C_{\text{b}})^l - C_{\text{in}}^l. \quad (5.22)$$

For the proof we first need the following new theory. We introduced the triangle condition as an indicator for mean-field behavior. Another related condition is the *open triangle condition*, stating that

$$\lim_{n \rightarrow \infty} \sup \{\tau_{p_c}^{*3}(x) \mid x \in \mathbb{Z}^d, |x| \geq n\} = 0, \quad (5.23)$$

where we recall  $\tau_{p_c}^{*3}(x)$  as the open triangle at criticality from (3.18). A lot of times the open triangle condition is more useful than its counterpart. The following theorem gives us a practical relationship between these two conditions, so we can use whatever fits better.

**Theorem 5.7** (Equivalent triangle conditions [6, 33]). *The open triangle condition holds if and only if the triangle condition holds.*

We note that the statement also holds for any transitive graph. The fact that the open triangle condition implies the (closed) triangle condition is not difficult to see, if we use repeatedly that if 0 and  $e$  are neighbors, there is a constant  $c > 0$  that is independent of 0,  $e$  and  $v$  such that  $c\mathbb{P}_{p_c}(v \longleftrightarrow e) \geq \mathbb{P}_{p_c}(v \longleftrightarrow 0)$ . This claim can be shown by an easy FKG argument. The proof of the other direction is more advanced and Kozma used operator theory of spectral theory to show it [33]. The statement for  $\mathbb{Z}^d$  is older and was shown with fourier transformations by Barsky and Aizenman [6]. The heavy usage of fourier transformations in high dimensions is typical for results on  $\mathbb{Z}^d$ .

*Proof of Proposition 5.6.* If the triangle condition holds, we have by Theorem 5.7 that also the open triangle condition is fulfilled. Therefore, for every  $\epsilon > 0$  there exist some  $r < \infty$  such that if  $x, y \in \mathbb{Z}^d$  have at least distance  $r$  we have  $\tau_{p_c}^{*3}(x, y) \leq \epsilon$ . This means there exist  $K(l, d)$  such that if  $x$  and  $y$  have at least distance  $K(l, d)$  we can receive the bound

$$\tau_{p_c}^{*2}(x, y) \leq \tau_{p_c}^{*3}(x, y) \leq \frac{p_c^{2+l}}{16 d l^2}, \quad (5.24)$$

where we chose  $\epsilon$  to the right side of (5.24). If we combine Theorem 5.3 and Theorem 5.4

with  $p = p_c$  and  $v_1, \dots, v_l \in \mathbb{Z}^d$  with at least distance  $K(l, d)$  we get

$$\begin{aligned}
& \mathbb{P}_{p_c, \mathbf{h}}(\mathcal{C}(v_1), \dots, \mathcal{C}(v_l) \text{ are disjoint and } v_i \longleftrightarrow \mathcal{G}_i \text{ for every } 1 \leq i \leq n) \\
& \geq \prod_{i=1}^l M_{p_c, h_i} - \frac{8d}{p_c^2} \binom{l-1}{2} \left( \prod_{i=1}^l M_{p_c, h_i} \right) \sup_{1 \leq i < j \leq l} \tau_{p_c}^{*2}(v_i, v_j) \\
& \quad - l(l-1) \left( \prod_{i=1}^l M_{p_c, h_i} \right) \sup_{1 \leq i < j \leq l} \tau_{p_c}^{*3}(v_i, v_j) \\
& \geq \prod_{i=1}^l M_{p_c, h_i} - \frac{4d}{p_c^2} (l-1)(l-2) \left( \prod_{i=1}^l M_{p_c, h_i} \right) \sup_{1 \leq i < j \leq l} \tau_p^{*3}(v_i, v_j) \\
& \quad - l(l-1) \left( \prod_{i=1}^l M_{p_c, h_i} \right) \sup_{1 \leq i < j \leq l} \tau_{p_c}^{*3}(v_i, v_j) \\
& = \left( \prod_{i=1}^l M_{p_c, h_i} \right) \left( 1 - \left( \frac{4d}{p_c^2} (l-1)(l-2) + l(l-1) \right) \sup_{1 \leq i < j \leq l} \tau_{p_c}^{*3}(v_i, v_j) \right) \\
& \geq \frac{1}{2} \prod_{i=1}^l M_{p_c, h_i}.
\end{aligned} \tag{5.25}$$

Fix now  $\mathbf{h} = (1/n^2, \dots, 1/n^2)$ . Applying Theorem 5.2 here then yields

$$\mathbb{P}_{p_c, \mathbf{h}}(\mathcal{C}(v_1), \dots, \mathcal{C}(v_l) \text{ are disjoint and } v_i \longleftrightarrow \mathcal{G}_i \text{ for every } 1 \leq i \leq n) \geq \frac{1}{2K^{l/2}} \frac{1}{n^l}. \tag{5.26}$$

We first show the proof for  $l = 2$ . We do this to get a good grasp of the strategy and the generalization to arbitrary  $l \geq 2$  is clearer. Our goal is to find a corresponding upper bound to (5.26) without the appearance of any ghost field. Instead we want an estimation using intrinsic arms. For that, we write  $\mathcal{A}$  for the event that  $\mathcal{C}(v_1)$  and  $\mathcal{C}(v_2)$  are disjoint. We can rephrase the magnetizations as an expectation and split the events according to their intrinsic radius:

$$\begin{aligned}
& \mathbb{P}_{p_c, \mathbf{h}}(\mathcal{C}(v_1) \text{ and } \mathcal{C}(v_2) \text{ are disjoint and } v_i \longleftrightarrow \mathcal{G}_i \text{ for } i = 1, 2) \\
& = \mathbb{E}_{p_c} \left[ \left( 1 - e^{-\frac{1}{n^2} |\mathcal{C}(v_1)|} \right) \left( 1 - e^{-\frac{1}{n^2} |\mathcal{C}(v_2)|} \right) \mathbb{1}(\mathcal{A}) \right] \\
& = \mathbb{E}_{p_c} \left[ \left( 1 - e^{-\frac{1}{n^2} |\mathcal{C}(v_1)|} \right) \left( 1 - e^{-\frac{1}{n^2} |\mathcal{C}(v_2)|} \right) \mathbb{1}(\mathcal{A}) \mathbb{1}(\text{rad}_{\text{int}}(v_1) \leq n \text{ and } \text{rad}_{\text{int}}(v_2) \leq n) \right] \\
& \quad + \mathbb{E}_{p_c} \left[ \left( 1 - e^{-\frac{1}{n^2} |\mathcal{C}(v_1)|} \right) \left( 1 - e^{-\frac{1}{n^2} |\mathcal{C}(v_2)|} \right) \mathbb{1}(\mathcal{A}) \mathbb{1}(\text{rad}_{\text{int}}(v_1) > n \text{ and } \text{rad}_{\text{int}}(v_2) > n) \right] \\
& \quad + 2 \mathbb{E}_{p_c} \left[ \left( 1 - e^{-\frac{1}{n^2} |\mathcal{C}(v_1)|} \right) \left( 1 - e^{-\frac{1}{n^2} |\mathcal{C}(v_2)|} \right) \mathbb{1}(\mathcal{A}) \mathbb{1}(\text{rad}_{\text{int}}(v_1) \leq n \text{ and } \text{rad}_{\text{int}}(v_2) > n) \right].
\end{aligned} \tag{5.27}$$

We now bound the three terms on the right hand side of (5.27). Note that the condition  $\text{rad}_{\text{int}}(v_1) \leq n$  implies that  $\mathcal{C}(v_1) = B(v_1, n)$ , where  $B(v_1, n)$  denotes the intrinsic ball around  $v_1$  with radius  $n$ . If we use the well know bound  $1 - e^{-x} \leq x$  we can estimate the

first term by

$$\begin{aligned}
& \mathbb{E}_{p_c} \left[ \left( 1 - e^{-\frac{1}{n^2} |\mathcal{C}(v_1)|} \right) \left( 1 - e^{-\frac{1}{n^2} |\mathcal{C}(v_2)|} \right) \mathbb{1}(\mathcal{A}) \mathbb{1}(\text{rad}_{\text{int}}(v_1) \leq n \text{ and } \text{rad}_{\text{int}}(v_2) \leq n) \right] \\
& \leq \frac{1}{n^4} \mathbb{E}_{p_c} \left[ |\mathcal{C}(v_1)| |\mathcal{C}(v_2)| \mathbb{1}(\mathcal{A}) \mathbb{1}(\text{rad}_{\text{int}}(v_1) \leq n \text{ and } \text{rad}_{\text{int}}(v_2) \leq n) \right] \\
& \leq \frac{1}{n^4} \sum_k \sum_l k l \mathbb{P}_{p_c}(\{|B(v_1, n)| = l\} \cap \{|B(v_2, n)| = k\} \cap \mathcal{A}) \\
& \leq \frac{1}{n^4} \sum_k \sum_l k l \mathbb{P}_{p_c}(\{|B(v_1, n)| = l\} \circ \{|B(v_2, n)| = k\}) \\
& \leq \frac{1}{n^4} \sum_k \sum_l k l \mathbb{P}_{p_c}(\{|B(v_1, n)| = l\}) \mathbb{P}_{p_c}(\{|B(v_2, n)| = k\}) \\
& = \frac{1}{n^4} \mathbb{E}_{p_c} \left[ |B(v_1, n)| \right] \mathbb{E}_{p_c} \left[ |B(v_2, n)| \right] \leq \frac{C_b^2}{n^2},
\end{aligned} \tag{5.28}$$

where  $C_b$  is the constant from the upper bound to the expected intrinsic volume. For the second term we can write

$$\begin{aligned}
& \mathbb{E}_{p_c} \left[ \left( 1 - e^{-\frac{1}{n^2} |\mathcal{C}(v_1)|} \right) \left( 1 - e^{-\frac{1}{n^2} |\mathcal{C}(v_2)|} \right) \mathbb{1}(\mathcal{A}) \mathbb{1}(\text{rad}_{\text{int}}(v_1) > n \text{ and } \text{rad}_{\text{int}}(v_2) > n) \right] \\
& \leq \mathbb{E}_{p_c} \left[ \mathbb{1}(\mathcal{A}) \mathbb{1}(\text{rad}_{\text{int}}(v_1) > n \text{ and } \text{rad}_{\text{int}}(v_2) > n) \right] \\
& = \mathbb{P}_{p_c}(\{\text{rad}_{\text{int}}(v_1) > n\} \cap \{\text{rad}_{\text{int}}(v_2) > n\} \cap \mathcal{A}) \\
& \leq \mathbb{P}_{p_c}(\{\text{rad}_{\text{int}}(v_1) \geq n\} \circ \{\text{rad}_{\text{int}}(v_2) \geq n\}).
\end{aligned} \tag{5.29}$$

The last part can be bounded by

$$\begin{aligned}
& \mathbb{E}_{p_c} \left[ \left( 1 - e^{-\frac{1}{n^2} |\mathcal{C}(v_1)|} \right) \left( 1 - e^{-\frac{1}{n^2} |\mathcal{C}(v_2)|} \right) \mathbb{1}(\mathcal{A}) \mathbb{1}(\text{rad}_{\text{int}}(v_1) \leq n \text{ and } \text{rad}_{\text{int}}(v_2) > n) \right] \\
& \leq \frac{1}{n^2} \mathbb{E}_{p_c} \left[ |B(v_1, n)| \mathbb{1}(\mathcal{A}) \mathbb{1}(\text{rad}_{\text{int}}(v_2) > n) \right] \\
& \leq \frac{1}{n^2} \sum_k k \mathbb{P}_{p_c}(\{|B(v_1, n)| = k\} \cap \mathcal{A} \cap \{\text{rad}_{\text{int}}(v_2) > n\}) \\
& \leq \frac{1}{n^2} \sum_k k \mathbb{P}_{p_c}(\{|B(v_1, n)| = k\} \circ \{\text{rad}_{\text{int}}(v_2) > n\}) \\
& \leq \frac{1}{n^2} \sum_k k \mathbb{P}_{p_c}(|B(v_1, n)| = k) \mathbb{P}_{p_c}(\text{rad}_{\text{int}}(v_2) > n) \\
& = \frac{1}{n^2} \mathbb{E}_{p_c} \left[ |B(v_1, n)| \right] \mathbb{P}_{p_c}(\text{rad}_{\text{int}}(v_2) > n) \\
& \leq \frac{C_b C_{\text{in}}}{n^2}.
\end{aligned} \tag{5.30}$$

Combining these estimates we get

$$\begin{aligned} & \mathbb{P}_{p_c, \mathbf{h}}(\mathcal{C}(v_1) \text{ and } \mathcal{C}(v_2) \text{ are disjoint and } v_i \longleftrightarrow \mathcal{G}_i \text{ for } i = 1, 2) \\ & \leq \mathbb{P}_{p_c}(\{\text{rad}_{\text{int}}(v_1) \geq n\} \circ \{\text{rad}_{\text{int}}(v_2) \geq n\}) + \frac{C_b^2 + 2C_b C_{\text{in}}}{n^2}. \end{aligned} \quad (5.31)$$

This means that if we use (5.26) we can conclude that

$$\mathbb{P}_{p_c}(\{\text{rad}_{\text{int}}(v_1) \geq n\} \circ \{\text{rad}_{\text{int}}(v_2) \geq n\}) \geq \frac{1}{n^2} \left( \frac{1}{2K} - C_b^2 - 2C_b C_{\text{in}} \right). \quad (5.32)$$

Unsurprisingly, the approach for  $l \geq 2$  is the same, but more laborious. We again choose  $\mathbf{h} = (1/n^2, \dots, 1/n^2)$  and like before compute

$$\begin{aligned} & \mathbb{P}_{p_c, \mathbf{h}}(\mathcal{C}(v_1), \dots, \mathcal{C}(v_l) \text{ are disjoint and } v_i \longleftrightarrow \mathcal{G}_i \text{ for } i = 1, \dots, l) \\ & = \mathbb{E}_{p_c} \left[ \prod_{i=1}^l \left( 1 - e^{-\frac{1}{n^2} |\mathcal{C}(v_i)|} \right) \mathbb{1}(A) \right] \\ & = \mathbb{E}_{p_c} \left[ \prod_{i=1}^l \left( 1 - e^{-\frac{1}{n^2} |\mathcal{C}(v_i)|} \right) \left( \mathbb{1}(\text{rad}_{\text{int}}(v_i) \geq n) + \mathbb{1}(\text{rad}_{\text{int}}(v_i) < n) \right) \mathbb{1}(A) \right] \\ & \leq \mathbb{E}_{p_c} \left[ \prod_{i=1}^l \left( \mathbb{1}(\text{rad}_{\text{int}}(v_i) \geq n) + \frac{|B(v_i, n)|}{n^2} \right) \mathbb{1}(A) \right] \\ & = \sum_{S \subseteq \{1, \dots, l\}} \mathbb{E}_{p_c} \left[ \prod_{i \in S} \mathbb{1}(\text{rad}_{\text{int}}(v_i) \geq n) \prod_{k \in S^c} \frac{|B(v_k, n)|}{n^2} \mathbb{1}(A) \right] \\ & \leq \mathbb{P}_{p_c}(\{\text{rad}_{\text{int}}(v_1) \geq n\} \circ \dots \circ \{\text{rad}_{\text{int}}(v_l) \geq n\}) \\ & \quad + \sum_{S \subseteq \{1, \dots, l\}} \mathbb{E}_{p_c} \left[ \prod_{i \in S} \mathbb{1}(\text{rad}_{\text{int}}(v_i) \geq n) \prod_{k \in S^c} \frac{|B(v_k, n)|}{n^2} \mathbb{1}(A) \right]. \end{aligned} \quad (5.33)$$

With the same procedure as before we use the BKR inequality to receive

$$\begin{aligned} & \sum_{S \subseteq \{1, \dots, l\}} \mathbb{E}_{p_c} \left[ \prod_{i \in S} \mathbb{1}(\text{rad}_{\text{int}}(v_i) \geq n) \prod_{k \in S^c} \frac{|B(v_k, n)|}{n^2} \mathbb{1}(A) \right] \\ & \leq \sum_{S \subseteq \{1, \dots, l\}} \mathbb{P}_{p_c}(\text{rad}_{\text{int}} \geq n)^{|S|} \left( \frac{\mathbb{E}_{p_c} |B(n)|}{n^2} \right)^{|S^c|} \\ & = \sum_{m=0}^{l-1} \binom{l}{m} \mathbb{P}_{p_c}(\text{rad}_{\text{int}} \geq n)^m \left( \frac{\mathbb{E}_{p_c} |B(n)|}{n^2} \right)^{l-m} \\ & \leq \sum_{m=0}^{l-1} \binom{l}{m} \left( \frac{C_{\text{in}}}{n} \right)^m \left( \frac{C_b}{n} \right)^{l-m} = \frac{1}{n^l} \left( \sum_{m=0}^{l-1} \binom{l}{m} C_{\text{in}}^m C_b^{l-m} \right). \end{aligned} \quad (5.34)$$

Combining these estimates with (5.26) we conclude

$$\frac{1}{n^l} \left( \frac{1}{2K^{l/2}} - G \right) \leq \mathbb{P}_{p_c}(\{\text{rad}_{\text{int}}(v_1) \geq n\} \circ \dots \circ \{\text{rad}_{\text{int}}(v_l) \geq n\}), \quad (5.35)$$

finishing our proof.  $\square$

An immense amount of theory was necessary to get to our statement. We used the one-arm exponent and the growth of the intrinsic balls, established in Theorem 4.3 for which we needed the critical exponents  $\delta, \gamma, \eta$ , or even  $\beta$ , if we use our coupling strategy. For these exponents to exist we needed the triangle condition to hold, which is far from trivial to show.

What would be left to check is that the constant of the right hand side of (5.32) is actually positive. We hope that this is the case, at least for  $d$  high enough. This would then show our conjecture, under the condition that  $d$  is sufficiently large. On the other hand, a lower bound on a probability by a negative number would not be very impressive. First we look at the behavior of  $K$  with respect to  $d$ . For that, we use (3.1) to control  $p_c$ , yielding

$$K = 1 + \frac{2dp_c}{1-p_c} \geq 1 + \frac{1}{1-p_c} \geq 2. \quad (5.36)$$

Next, we have to check how  $C_b$  behaves if we increase the dimension. Since we subtract terms with  $C_b$ , we would like that constant to be sufficiently low. If we go back to the proof of the one-arm exponent (4.51), we see that  $C_{\text{in}}$  is actually just a natural number, independent of any parameters in question. We had  $C_b = 4C_\gamma/p_c$ , where  $C_\gamma$  is the upper constant in Theorem 3.4. If  $\Delta_{p_c}(e)$  is the open triangle and  $e \sim 0$ , Lemma 3.2 yields

$$\Delta_{p_c}(e) \approx \Delta_{p_c} - 1 \leq \frac{C_{p_c}}{d}, \quad (5.37)$$

and thus the open triangle can be made small, if we choose  $d$  big enough and the explicit upper bound from (3.23) holds. So

$$C_b = \frac{4C_\gamma}{p_c} = \frac{4}{2d(1-\Delta(e))p_c} \xrightarrow{d \rightarrow \infty} 4. \quad (5.38)$$

However this does not give us the desired lower constant for our conjecture, since the large value for  $C_{\text{in}}$  implies that even for high dimensions

$$\frac{1}{2K^{l/2}} - G < 0. \quad (5.39)$$

Obviously, bounding a probability by a negative number from below is banal and not enough for  $\rho_l = 1/l$ . We needed that  $C_b$  converges to 0 as  $d$  goes to infinity. This however cannot hold, since it would imply that  $\mathbb{E}_{p_c}|B(n)| \rightarrow 0$  for large  $d$ .

Thus the conjecture remains open and we can only rigorously give the mean-field bounds  $1/(2l) \leq \rho_l \leq 1/l$ .

### 5.2.1 Different Strategies

We want to point out that the two lemmas our proof relied on were used differently by Hutchcroft when he studied multiple arms in the nonamenable setting for Theorem 5.1. We chose  $\mathbf{h}$  in a way to end up with the correct exponent on  $n$ . In [29] a limit argument was used, that does not go through for our amenable setting. One can show that under the  $L^2$  boundedness condition, (5.25) also holds in the slightly supercritical regime. We then can use this inequality together with  $\mathbf{h} \searrow 0$  and  $\beta = 1$  to receive for  $p_c < p$

$$\mathbb{P}_p(v_1, \dots, v_l \text{ are in disjoint infinite clusters}) \asymp (p - p_c)^l. \quad (5.40)$$

Even though this result is surely interesting for a model where multiple disjoint clusters are allowed, the uniqueness of the infinite cluster for percolation on  $\mathbb{Z}^d$  tells us that this will lead us nowhere.

We will briefly discuss and present another proof strategy that we think does not work out. Yet we saw this procedure a lot of times in similar settings, so we think it is beneficial to present the method. One natural thing one might want to do is trying to adapt the way Kozma and Nachmias were able to get from one arm to multiple arms in the extrinsic setting.

In contrast to the intrinsic setting the events  $\{0 \longleftrightarrow \partial\Lambda_n\}$  are increasing. Indeed we do not have the problem of creating a shorter path here. Therefore we can even use the original version of the BK inequality. Since the upper bound is an immediate consequence of the BK inequality the whole trouble again is finding the lower bound and therefore we need some sort of “inverse BK method” to get a bound in the other direction. The proofs of both Theorem 4.11 and Theorem 5.4 are heavily inspired by the techniques used in the proof of the BK inequality for increasing events [18, Section 2.5]. For this we want to consider not just one edge  $e \in \mathcal{E}(\mathbb{Z}^d)$  connecting two adjacent vertices, but two parallel edges  $e'$  and  $e''$ , who are open independently from each other. After this we will successively merge edges and try to measure the changed probability of our events.

To model this more formally we follow Hutchcroft’s notation and let  $\omega_0$  and  $\omega_\infty$  be independent copies of Bernoulli- $p$  bond percolation and let  $\mathbb{P}_\otimes$  be their joint law. Next we enumerate the edge set  $\mathcal{E}(\mathbb{Z}^d)$  by  $e_1, e_2, \dots$ . Then, for each  $m \geq 0$  we can merge the two configurations  $\omega_0$  and  $\omega_\infty$  and define  $\omega_m$  by

$$\omega_m(e_i) = \begin{cases} \omega_\infty(e_i) & \text{if } i \leq m, \\ \omega_0(e_i) & \text{if } i > m. \end{cases} \quad (5.41)$$

For an event  $\mathcal{A} \in \mathcal{F}$  and each  $m \in \{0, 1, \dots\} \cup \{\infty\}$ , denote by  $\mathcal{A}_m$  the event that  $\omega_m$  satisfies  $\mathcal{A}$ . Given another event  $\mathcal{B}$  this yields

$$\mathbb{P}_p(\mathcal{A} \circ \mathcal{B}) = \mathbb{P}_\otimes(\mathcal{A}_0 \circ \mathcal{B}_0). \quad (5.42)$$

Likewise, we can also deduce

$$\mathbb{P}_p(\mathcal{A})\mathbb{P}_p(\mathcal{B}) = \mathbb{P}_\otimes(\mathcal{A}_0 \circ \mathcal{B}_\infty) = \lim_{m \rightarrow \infty} \mathbb{P}_\otimes(\mathcal{A}_0 \circ \mathcal{B}_m). \quad (5.43)$$

As a side note: If we would now fix  $\mathcal{A}$  and  $\mathcal{B}$  to be increasing, the classical BK inequality would follow from (5.42) and (5.43) if we could show that for all  $m \geq 0$

$$\mathbb{P}_\otimes(\mathcal{A}_0 \circ \mathcal{B}_m) \geq \mathbb{P}_\otimes(\mathcal{A}_0 \circ \mathcal{B}_{m-1}). \quad (5.44)$$

We continue by noticing that from (5.42) and (5.43) we can follow

$$\sum_{m=1}^{\infty} [\mathbb{P}_\otimes(\mathcal{A}_0 \circ \mathcal{B}_m) - \mathbb{P}_\otimes(\mathcal{A}_0 \circ \mathcal{B}_{m-1})] = \mathbb{P}_p(\mathcal{A})\mathbb{P}_p(\mathcal{B}) - \mathbb{P}_p(\mathcal{A} \circ \mathcal{B}). \quad (5.45)$$

We could now set  $p = p_c$ ,  $\mathcal{A} = \{\text{rad}_{\text{int}}(v_1) \geq n\} \circ \dots \circ \{\text{rad}_{\text{int}}(v_{l-1}) \geq n\}$  and  $\mathcal{B} = \{\text{rad}_{\text{int}}(v_l) \geq n\}$ . Write  $e_m = \{e_m^-, e_m^+\}$ . If we can prove that there is a constant  $A > 0$

that is independent from  $n$ , such that for all  $m \geq 0$

$$\mathbb{P}_{\otimes}(\mathcal{A}_0 \circ \mathcal{B}_m) - \mathbb{P}_{\otimes}(\mathcal{A}_0 \circ \mathcal{B}_{m-1}) \leq A \sum_{j=1}^{l-1} \mathbb{P}_{p_c}(v_l \longleftrightarrow e_m^-) \mathbb{P}_{p_c}(v_j \longleftrightarrow e_m^-) \left( \mathbb{P}_{p_c}(\text{rad}_{\text{int}} \geq n) \right)^l, \quad (5.46)$$

we can use induction on  $l$ , remember what we know about  $\rho_{\text{in}}$  and (5.45) to compute

$$\begin{aligned} \mathbb{P}_p(\mathcal{A} \circ \mathcal{B}) &\geq \frac{c_{\text{in}} c_{l-1}}{n^l} - \frac{AC_{\text{in}}^l}{n^l} \sum_{m=1}^{\infty} \sum_{j=1}^{l-1} \mathbb{P}_{p_c}(v_l \longleftrightarrow e_m^-) \mathbb{P}_{p_c}(v_j \longleftrightarrow e_m^-) \\ &\geq \frac{c_{\text{in}} c_{l-1}}{n^l} - \frac{AC_{\text{in}}^l (l-1)}{n^l} \sup_{1 \leq i < j \leq l} \tau_{p_c}^{*2}(v_i, v_j) \\ &\geq \left( c_{\text{in}} c_{l-1} - AC_{\text{in}}^l (l-1) \sup_{1 \leq i < j \leq l} \tau_{p_c}^{*2}(v_i, v_j) \right) \frac{1}{n^l}. \end{aligned} \quad (5.47)$$

So we are in the same boat as before: We have found a lower bound and need to check that the constant is positive, making the bound meaningful. But now we could actually do it since we have control over  $\tau_{p_c}^{*2}$ : Since the triangle condition holds, we have by Theorem 5.7 that also the open triangle condition is fulfilled. Therefore for every  $\epsilon > 0$  there exist some  $r < \infty$  such that if  $x, y \in \mathbb{Z}^d$  have at least distance  $r$  we have  $\tau_{p_c}^{*3}(x, y) < \epsilon$ . This means there exist  $K(l, d)$  such that if  $x$  and  $y$  have at least distance  $K(l, d)$  we can receive the bound

$$\tau_{p_c}^{*2}(x, y) \leq \tau_{p_c}^{*3}(x, y) < \frac{c_{\text{in}} c_{l-1}}{2A(l-1)C_{\text{in}}^l}, \quad (5.48)$$

so we would be finished if we set  $v_1, \dots, v_l$  at distance  $K(l, d)$ .

Unfortunately, to derive an upper bound like (5.46), we would normally like to have increasing events. We see that for all increasing sets  $\mathcal{A}$  and  $\mathcal{B}$  we must have  $\omega_{\infty}(e_m) = 1$  on  $\mathcal{A}_0 \circ \mathcal{B}_m \setminus \mathcal{A}_0 \circ \mathcal{B}_{m-1}$ . Then the proof of the BK inequality constructs a measure preserving bijection (meaning the sets have the same probability) between

$$\mathcal{A}_0 \circ \mathcal{B}_m \setminus \mathcal{A}_0 \circ \mathcal{B}_{m-1} \text{ holds, } \omega_0(e_m) = 0 \text{ and } \omega_{\infty}(e_m) = 1 \quad (5.49)$$

and

$$\mathcal{A}_0 \circ \mathcal{B}_{m-1} \setminus \mathcal{A}_0 \circ \mathcal{B}_m \text{ holds, } \omega_0(e_m) = 1 \text{ and } \omega_{\infty}(e_m) = 0. \quad (5.50)$$

Using these observations we can write the probability difference we wanted to control quite nicely as

$$\begin{aligned} \mathbb{P}_{\otimes}(\mathcal{A}_0 \circ \mathcal{B}_m) - \mathbb{P}_{\otimes}(\mathcal{A}_0 \circ \mathcal{B}_{m-1}) &= \mathbb{P}_{\otimes}(\mathcal{A}_0 \circ \mathcal{B}_m \setminus \mathcal{A}_0 \circ \mathcal{B}_{m-1}) - \mathbb{P}_{\otimes}(\mathcal{A}_0 \circ \mathcal{B}_{m-1} \setminus \mathcal{A}_0 \circ \mathcal{B}_m) \\ &\leq \mathbb{P}_{\otimes}(\mathcal{A}_0 \circ \mathcal{B}_m \setminus \mathcal{A}_0 \circ \mathcal{B}_{m-1}, \omega_0(e_m) = \omega_{\infty}(e_m) = 1), \end{aligned} \quad (5.51)$$

and then continue from there. However we already know that intrinsic arm events are not increasing, so we cannot easily proceed like this.

# Chapter 6

## Conclusion

We now quickly recapitulate what we learned in this thesis. After we first got to know the basics of percolation theory and the central results plus inequalities, we saw that percolation on  $\mathbb{Z}^d$  for  $d > 6$  behaves similar to percolation on the regular tree, or like a branching random walk. For that we defined numerous critical exponents that should describe vital objects that express properties of the model. We saw that these critical exponents take on their mean-field value under the triangle condition and that the exact values are the same on a regular tree.

Then we studied arm events and showed that  $\rho_{\text{in}} = 1$  on both  $\mathbb{T}_{2d}$  and the hypercubic lattice. We now know that even on very general graphs we can give similar estimates concerning intrinsic radii. We then tried to generalize this to the occurrence of multiple disjoint geodesic paths of a certain length. Here the work of Hutchcroft was presented, who showed that the BKR inequality is sharp in the case for nonamenable graphs. Under the philosophy of mean-field behavior we conjectured that the same holds for our amenable setting of  $\mathbb{Z}^d$  and gave the mean-field bounds  $1/(2l) \leq \rho_l \leq 1/l$ . Two proof ideas were given and discussed: a magnetization approach and an inverse BK method.

After going through everything, the reader hopefully saw an overview of the percolation basics in high dimensions, got to learn how to use critical exponents and saw numerous proof methods of probability theory, like couplings or the second moment method. Throughout, numerous literature references were given for further exploration of the theory. What remains open in the full generality is the missing lower bound for  $\rho_l = 1/l$  to hold. To tackle this, one would have to come up with new ideas.

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## Erklärung der Urheberschaft

Ich erkläre hiermit, dass ich die vorliegende Arbeit ohne Hilfe Dritter und ohne Benutzung anderer als der angegebenen Hilfsmittel angefertigt habe; die aus fremden Quellen übernommenen Gedanken sind als solche kenntlich gemacht.

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